

RESEARCH ARTICLE SUMMARY

AGING

Restored clearance of senescent neutrophils by tissue-resident macrophages limits organ aging

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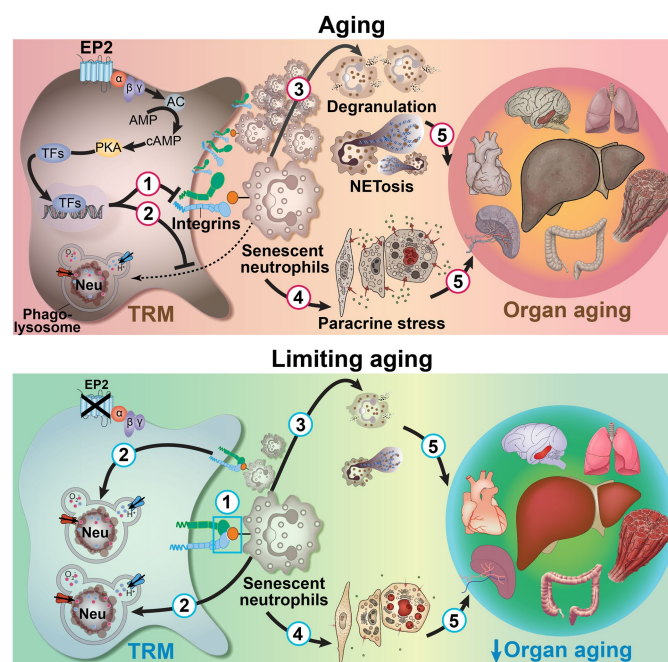
Full article and list of author affiliations: <https://doi.org/10.1126/science.aea3075>

INTRODUCTION: Aging is accompanied by parallel functional decline across organs, but the cellular drivers remain unclear. Tissue-resident macrophages (TRMs), long-lived cells that comprise 60 to 90% of macrophages in major organs, maintain homeostasis through efferocytosis of apoptotic and senescent cells. Neutrophils, the most abundantly produced and shortest-lived leukocytes (more than 100 billion generated daily in humans), require continuous TRM clearance; uncleared aged neutrophils release proteases and extracellular traps that damage tissues and propagate aging. TRMs express the prostaglandin E₂ (PGE₂) receptor EP2, which suppresses macrophage metabolism and phagocytosis in aging. Whether impaired TRM efferocytosis drives the accumulation of senescent neutrophils that promote organ aging and whether inhibition of EP2 signaling can restore this process, remain unknown.

RATIONALE: We studied aged mice in which EP2 signaling on TRMs was selectively reduced, either genetically (TRM-specific EP2 deletion) or pharmacologically, to define how TRM dysfunction shapes organ-wide aging. TRMs are long-lived gatekeepers of tissue homeostasis, and EP2 offers a tractable target because its activity increases in aged macrophages and suppresses their metabolic and phagocytic function. Using complementary genetic and pharmacological approaches, we tested whether restoring TRM function reverses organ aging and identified which efferocytic substrate is most affected. The same design also defined the molecular step at which EP2 acts within TRMs and extended our findings to aged human tissues.

RESULTS: In aged mice, TRM-specific EP2 deletion restored mitochondrial fitness and immune homeostasis, and reversed cognitive decline, frailty, sarcopenia, adiposity, and cardiac dysfunction toward youthful states. Plasma proteomics identified the liver as a major source of age-associated immune changes. Single-cell RNA-seq of mouse liver and multiorgan flow cytometry revealed accumulation of senescent CXCR4⁺ neutrophils across efferocytic organs in aging. These cells exhibited the senescence-associated secretory phenotype (SASP), DNA damage response activation, cell cycle inhibitor induction, NETosis, and anti-apoptosis programs, and were efficiently cleared following EP2 deletion. Liver multiplex imaging localized paracrine stress to parenchymal cells neighboring senescent neutrophils. Ex vivo efferocytosis assays showed that aged TRMs were most impaired in clearing senescent neutrophils relative to apoptotic substrates, with both functions restored by EP2 deletion or pharmacologic antagonism. Mechanistically, EP2 signaling suppressed integrin-dependent stabilization of senescent neutrophils on TRMs and downstream engulfment. Analyses of human liver and heart datasets revealed conserved EP2 up-regulation in aged TRMs, enrichment of senescent neutrophils, and reduced TRM-neutrophil interactions.

CONCLUSION: This work identifies EP2 signaling in TRMs as a central regulator of organ-wide aging through its control of



Tissue-resident macrophage (TRM) clearance of senescent neutrophils is restored by EP2 deletion or inhibition to limit organ aging. (Top) In aged TRMs, EP2 signaling suppresses (1) integrin engagement with senescent neutrophil (Neu) ligands required for clearance and (2) engulfment of senescent neutrophils into phagolysosomes, allowing senescent neutrophils to accumulate, (3) undergo degranulation and NETosis, and (4) exert paracrine stress on neighboring parenchymal cells, collectively driving organ aging. (Bottom) EP2 deletion or inhibition restores (1) integrin-mediated interaction and (2) phagolysosomal engulfment, limiting (3) degranulation and NETosis, (4) paracrine stress, and (5) organ aging. NETosis, neutrophil extracellular trap (NET) formation; AC, adenylyl cyclase; PKA, protein kinase A; TFs, transcription factors.

senescent neutrophil clearance, reframing aging as a failure of active cellular clearance rather than passive degeneration. With age, neutrophils acquire senescence-associated features, and their accumulation drives tissue injury through two converging mechanisms: intrinsic degranulation and NETosis, and extrinsic paracrine stress on neighboring parenchymal cells. Pharmacological inhibition of EP2 restores TRM efferocytic capacity and promotes clearance of senescent neutrophils, positioning EP2 antagonism as a tractable therapeutic strategy for age-related organ and functional decline. □

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Restored clearance of senescent neutrophils by tissue-resident macrophages limits organ aging

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Aging disrupts tissue homeostasis across organ systems. Here, we identify tissue-resident macrophages (TRMs) as central coordinators of age-related organ decline through impaired clearance of senescent neutrophils, a process regulated by the immunomodulatory prostaglandin E2 (PGE₂) receptor EP2. Reducing TRM EP2 signaling in aged mice preserved youthful mitochondrial fitness and prevented cognitive decline, frailty, sarcopenia, adiposity, cardiac impairment, and systemic inflammation. Plasma proteomics implicated the liver as a major source of age-associated immune change, in which reduced TRM EP2 signaling rescued neutrophil efferocytosis and prevented paracrine stress in neighboring cells. Elevated TRM EP2 expression and senescent neutrophils were also observed in aged and diseased human tissues. Pharmacologic EP2 inhibition restored youthful neutrophil clearance, establishing impaired TRM efferocytosis as a reversible driver of organ decline in aging.

Aging is marked by the progressive breakdown of tissue integrity, regenerative capacity, and immune equilibrium (1, 2). This decline is tightly associated with development of systemic inflammation and multiple chronic disorders of aging, including cognitive decline, vascular disease, frailty, sarcopenia, and metabolic syndrome. Although molecular hallmarks of aging have been identified, the cellular events that initiate and propel tissue decline remain poorly defined (3).

Macrophages are central regulators of tissue homeostasis, inflammation, and repair (4). In aging, macrophage phenotypes shift toward chronic inflammation and phagocytic dysfunction, but whether these changes are causal or compensatory remains unresolved (5). Among macrophage subtypes, tissue-resident macrophages (TRMs) have drawn increasing attention as potential initiators of aging due to their early developmental origin, longevity, and integration into tissue niches (6, 7). TRMs arise largely from yolk sac progenitors and fetal liver monocytes, self-renew, and persist throughout life, with tissue-dependent postnatal contribution from monocyte-derived macrophages (MDMs), which are themselves short-lived with limited self-renewal (7–12).

TRMs comprise 60 to 90% of macrophages in the brain, liver, lungs, heart, and kidneys (7–9, 13, 14), and their long lifespan makes them particularly vulnerable to aging, as they accumulate metabolic, oxidative, and inflammatory injury over years to decades (15, 16). The brain contains almost exclusively TRMs, with an estimated 10 to 20 billion

microglia, while peripheral tissues collectively harbor approximately 200 billion TRMs (17, 18). Given their abundance and ubiquitous presence across organs, age-related changes in TRMs may represent an important cellular driver of organ dysfunction and systemic aging.

A core TRM function is efferocytosis, the clearance of apoptotic, senescent, and damaged cells that is essential for preventing chronic inflammation (19). Among primary TRM targets are neutrophils, the most abundantly produced immune cell, with more than 10 billion and 100 billion generated daily in mice and humans, respectively (20, 21). Neutrophils are among the shortest-lived immune cells, aging within hours in the circulation and requiring continuous clearance (22, 23). Uncleared aged neutrophils release proteases and extracellular traps that damage tissues, propagate inflammation, and promote aging (24–26), and are normally removed efficiently by TRMs in the liver, spleen, and bone marrow (27–30).

TRMs also express receptors for immunomodulatory signals, including those for the lipid messenger prostaglandin E2 (PGE₂), a downstream product of the inflammatory cyclooxygenase-2 (COX-2) pathway. PGE₂ elicits pro- or anti-inflammatory effects through four G protein-coupled receptors (EP1 to EP4). In aged macrophages, PGE₂-EP2 signaling suppresses glycolysis and mitochondrial respiration, disrupting phagocytosis and inflammation resolution (31). Here, we identify TRM clearance of senescent neutrophils as a critical driver of organ aging and show that this process is regulated by the PGE₂ receptor EP2.

Results

EP2 deficiency in aging TRMs preserves mitochondrial fitness and organ immune homeostasis

TRMs largely arise prenatally from yolk sac-derived erythro-myeloid progenitors (EMPs) and self-renew locally throughout life, whereas MDMs arise postnatally from bone marrow hematopoietic stem cells (HSCs) and are continuously replenished from the circulation, with tissue-dependent conversion of MDMs into long-tenured TRM-like cells (Fig. 1A) (8, 11, 12). CX3CR1 is highly expressed in TRMs but lower in MDMs, making it a practical marker for biased TRM targeting, which we further refined with residency-specific readouts. To examine EP2 signaling in TRMs, we crossed *Cx3cr1*^{CreER} mice with *Ptger2*^{lox/lox} mice to conditionally delete EP2 in CX3CR1⁺ cells after tamoxifen treatment (EP2 cKO; Fig. 1B).

We validated TRM-selective EP2 deletion in young (6 to 8 months old) wild-type (WT), (*Cx3cr1*^{CreER}), and EP2 cKO mice given tamoxifen 2 months earlier. In the liver, lungs, and heart, TRMs comprised 65 to 80% of tissue macrophages and expressed EP2 at >95%, with ~50 to 60% deletion in cKO mice. MDM EP2, detected only in the liver and lungs, was minimally reduced (<10%), due to low CX3CR1 expression and continuous MDM replenishment from CX3CR1[−] HSCs (32, 33) that dilutes Cre-mediated deletion, whereas TRM self-renewal keeps their deletion stable (fig. S1). Overall, the EP2⁺ macrophage fraction fell from ~80 to ~40% in cKO mice, driven almost entirely by TRMs.

Because CX3CR1 is expressed in both TRMs and MDMs, Cre-mediated deletion could in principle act on either population irrespective of tissue tenure. We therefore tested whether residency, rather than ontogeny, predicts EP2 deletion by stratifying liver and lung macrophages by TIM4, a canonical residency marker independent of ontogeny (34, 35). TIM4 was expressed in ~80 to 90% of liver and lung macrophages across young and aged mice of both genotypes, and TIM4⁺ cells coexpressed Clec4f (liver) and SiglecF (lungs) at ~85 to 95%, whereas TIM4[−] cells expressed these markers at <10%, establishing TIM4⁺ macrophages as bona fide TRMs. TIM4⁺ TRMs expressed EP2 at ~90 to 95% in WT and ~45 to 50% in cKO mice, whereas TIM4[−] macrophages maintained EP2 at ~25% regardless of genotype, with patterns preserved in aged mice (fig. S2). Thus, *Cx3cr1*^{CreER}-driven EP2 deletion segregates with residency: TIM4⁺ TRMs preferentially lose EP2, whereas TIM4[−] nonresident macrophages do not, indicating TRM-biased deletion independent of developmental origin.

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We next compared young and aged (23 to 25 months old) WT and EP2 cKO mice across the liver, lungs, and brain. In the brain, we further distinguished microglia (TRMs), damage-associated macrophages (DAMs), border-associated macrophages (BAMs), and MDMs (36, 37) (fig. S3A). TRMs predominated in lungs (~70 to 80%) and brain (>90%) regardless of age, whereas aged liver contained considerably fewer TRMs (~55%) than young liver (~75 to 80%), and this deficit was restored in aged EP2 cKO mice (Fig. 1C). CX3CR1 remained stably high in TRMs (~70 to 90%) and low in MDMs (~10 to 30%) across organs and ages (Fig. 1D). EP2 was most highly expressed in TRMs across all organs, with aged TRM deletion efficiency (~50 to 60%) matching that in young mice (Fig. 1E); in the brain, EP2 was also detected in DAMs (~70%), BAMs (~80%), and MDMs (~50%), with cKO reducing EP2 in TRMs, DAMs, and BAMs by ~40 to 50%, whereas MDM deletion remained negligible. Lymphoid EP2 was low in the liver and lungs (<5%) but elevated in meningeal innate lymphoid cells (ILCs), natural killer (NK) cells, and CD4 T cells (>40%), reflecting the distinct meningeal immune milieu (38). Because TRMs accounted for >95% of total EP2 loss, the overall fraction of EP2⁺ cells declined by ~50% across organs (Fig. 1, F and G), confirming that robust, TRM-biased EP2 deletion is maintained with aging.

To compare aging effects on TRMs versus MDMs, we examined the peritoneal cavity using established markers to distinguish TRMs (CD45⁺CD11b^{hi}F4/80⁺) from MDMs (CD45⁺CD11b^{int}F4/80⁺CCR2⁺) (39) (fig. S3B). In contrast to MDMs, TRM abundance declined considerably with age and this loss was prevented in aged EP2 cKO mice (Fig. 1H). This selective protection prompted us to examine mitochondrial function, a key determinant of TRM survival and phagocytic capacity (40). Aged TRMs showed reduced mitochondrial mass and membrane potential along with elevated superoxide, all of which were largely restored to youthful amounts in aged EP2 cKO mice (Fig. 1, I to K, and fig. S3, C to E). MDMs showed only increased superoxide with no changes in proportion, mass, or membrane potential.

Aging is accompanied by low-grade systemic inflammation underlying vascular, metabolic, and neurodegenerative disorders. We tested whether TRM EP2 deletion reverses this inflammatory state across plasma and major organs. In aged mice, TRM EP2 deletion reverted inflammatory profiles to youthful states across plasma, liver, colon, heart, kidneys, and hippocampus (Fig. 1, L to Q, and fig. S4, A to F). Plasma endotoxin was restored to near-youthful concentrations in aged EP2 cKO mice (Fig. 1R), reflecting improved gut barrier integrity or enhanced blood endotoxin clearance. Thus, TRMs drive systemic and organ-wide age-associated inflammation through PGE₂-EP2 signaling.

TRM EP2 deletion limits age-associated declines in organ function

Because TRM EP2 deletion restored youthful immune signatures across multiple organs and blood (Fig. 1), we tested whether these changes were accompanied by improvements in organ function.

Microglia, the TRMs of the brain, are yolk sac-derived and strongly express Cx3cr1 (41). To test whether microglia affect cognitive function, we examined young (6 to 8 months old) and aged (23 to 25 months old) male EP2 cKO and WT mice in the Barnes maze (hippocampal-dependent spatial memory) and novel object recognition (cortical-dependent recognition memory) tasks. Both short- and long-term spatial memory (days 4 and 14) were considerably improved in aged EP2 cKO relative to aged WT mice (Fig. 2A, and fig. S4, G and H). Aged EP2 cKO mice also regained object-based episodic memory, performing comparably to young mice (Fig. 2B). Thus, microglial EP2 signaling is an important driver of age-associated memory decline.

During routine handling, aged EP2 cKO mice appeared leaner and more physically fit than WT littermates. Whole-body MRI confirmed increased limb muscle volume and reduced visceral fat in aged EP2 cKO mice, approaching those of young mice (Fig. 2, C to E). These changes, along with decreased subcutaneous fat (fig. S4I), were associated with improved frailty scores (Fig. 2F), greater forelimb strength (Fig. 2G

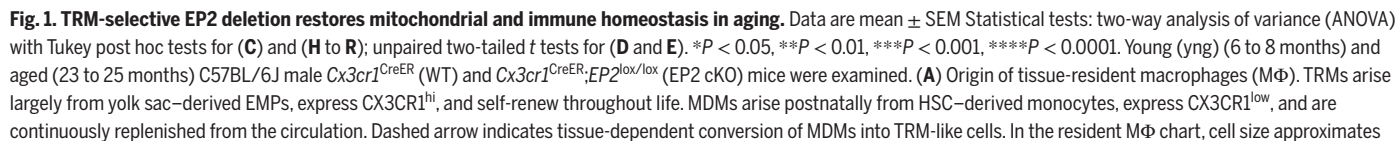
and fig. S4I), and preserved muscle fiber composition and architecture (Fig. 2, H and I). MHCIIa⁺ fibers, which confer endurance through oxidative metabolism (42), were substantially depleted in aged WT but restored in aged EP2 cKO mice. Aberrant myonuclear positioning, a hallmark of sarcopenia and impaired muscle regeneration (43), was also largely normalized toward a youthful pattern (Fig. 2J and fig. S4J).

We then tested whether the attenuation of skeletal muscle aging in aged EP2 cKO mice extends to the heart, a highly vascularized, mitochondria-rich organ whose function also deteriorates with age (44). Cardiac MRI revealed that TRM EP2 deletion preserved youthful left ventricular ejection fraction, septal wall thickness, and left ventricle (LV)/right ventricle (RV) ratio, consistent with improved systolic function (Fig. 2, K and L). Given the heart's high energy demands, we examined mitochondrial integrity in cardiomyocytes using transmission electron microscopy (TEM). Aged WT hearts had more lucent mitochondria with disrupted cristae; by contrast, EP2 cKO hearts showed more electron-dense mitochondria, similar to those of young hearts (Fig. 2, M and N, and fig. S4K). We also measured myocardial stiffness, which increases in aging and is driven in part by extracellular matrix deposition, particularly collagen. Trichrome staining revealed collagen accumulation in the ventricular wall in aged WT mice that was reduced to near-youthful amounts in aged EP2 cKO mice. Collagen-extracted and fiber-isolated views confirmed a reduction in fibrosis with preserved muscle fiber integrity in aged EP2 cKO mice (Fig. 2, O and P). Together, these findings indicate that TRM EP2 deletion limits detrimental structural and functional remodeling of the aging heart, and that selective targeting of EP2 in CX3CR1⁺ TRMs can broadly ameliorate age-associated declines in cognition, skeletal muscle, adiposity, and cardiac function.

The aging plasma proteome remains in a youthful state with TRM EP2 deletion

Given the broad, multiorgan improvements observed with TRM-specific EP2 deletion, we sought to identify systemic mediators of these changes. Because the blood circulation integrates immune and metabolic signals across tissues, we performed quantitative plasma proteomic profiling using tandem mass tag (TMT)-based liquid chromatography-tandem mass spectrometry (LC-MS/MS) in young and aged WT and EP2 cKO mice (Fig. 3A). After depleting high-abundance plasma proteins, 138 proteins were confidently quantified (>2 unique peptides). Of those, 71 were substantially altered with aging, and 59 were restored toward youthful amounts in aged EP2 cKO mice (Fig. 3B). Principal component analysis (PCA) revealed a clear separation of aged WT plasma from both aged EP2 cKO and young groups (Fig. 3C). Pathway enrichment of rescued proteins identified broad immune remodeling, including the acute phase response, IL-12 signaling, integrin signaling, IL-10 signaling, and neutrophil degranulation (Fig. 3, D to F).

To identify tissue sources driving these age-related plasma changes, we performed cell type enrichment analysis using tissue-specific proteomic signatures. This analysis revealed that hepatocytes, Kupffer cells (KCs), and cholangiocytes were the most enriched cellular sources (Fig. 3G), pointing to the liver as a major contributor to aging-related changes in blood. Although classically viewed as a metabolic hub, the liver is also one of the most TRM-rich organs (17), serving to filter portal and circulating blood while maintaining both hepatic and systemic immune homeostasis (45). Consistent with this, aging was associated with elevated serum biomarkers of liver inflammation and injury, including aspartate aminotransferase (AST), alanine aminotransferase (ALT), gamma-glutamyl transferase (GGT), alkaline phosphatase, and C-reactive protein, all of which were restored toward youthful amounts in aged EP2 cKO mice (Fig. 3H). These findings support a central role for the liver, and likely liver TRMs, in driving age-associated systemic immune changes that are largely prevented by TRM EP2 deletion.



relative abundance. LPM, large peritoneal macrophage; RPM, red pulp macrophage. **(B)** Generation of TRM-selective EP2 conditional knockout (EP2 cKO) mice. Tamoxifen was administered at 4 to 6 months of age to both genotypes. Young and aged mice were evaluated at 6 to 8 months and 23 to 25 months, respectively. i.p., intraperitoneal injection. **(C)** Mean TRM and MDM proportions within the macrophage pool in liver, lungs, and brain. $n = 8$ to 10 mice per group. **(D)** CX3CR1 expression percentage in TRM and MDM subsets across liver, lungs, and brain. $n = 18$ mice per age group (WT and cKO are combined). **(E)** EP2 expression percentage across immune cell subsets in the liver, lungs, and brain. Lowercase letters (a and b) denote significant differences between the wild type and cKO within each age group. Neu, neutrophil; ILC, innate lymphoid cell; DAM, damage-associated macrophage; BAM, border-associated macrophage. $n = 8$ to 10 mice per group. **(F)** Mean immune cell percentages per million CD45⁺ cells in the liver, lungs, and brain. Cell types are color coded as indicated. Data are averaged across all age and genotype groups. $N = 36$ mice per organ. **(G)** Mean EP2⁺ and EP2⁻ cell percentages per million CD45⁺ cells by immune subset in liver, lungs, and brain. Solid bars, EP2⁺; hatched bars, EP2⁻. Lymphoid (Lymph.) cells include ILC, NK, CD4 T, CD8 T, and B cells. Data are averaged within each genotype group combining young and aged mice. $n = 18$ mice per genotype. **(H)** Peritoneal TRM and MDM percentages within CD45⁺CD11b⁺ cells. PTM, peritoneal MΦ; $n = 6$ to 12 mice per group. **(I) to (K)** Mitochondrial features of peritoneal TRMs and MDMs: mitochondrial mass (MitoGreen MFI) (I), inner membrane potential ($\Delta\Psi$ m; MitoRed MFI) (J), and percentage of superoxide-producing MΦ (MitoSOX⁺) (K). MFI, mean fluorescence intensity. $n = 6$ to 12 mice per group. **(L to Q)** Unsupervised hierarchical clustering of significantly regulated immune factors ($*P < 0.05$, two-way ANOVA) in the plasma, liver, colon, heart, kidneys, and hippocampus by Luminex across young and aged WT and EP2 cKO mice. Each heatmap value represents the average of two technical replicates. $n = 7$ to 13 mice per group. **(R)** Plasma endotoxin in young and aged WT and EP2 cKO mice. $n = 7$ to 13 mice per group.

Senescent neutrophil accumulation across efferocytic organs in aging is reversed by TRM EP2 deletion

Given the liver's central role in shaping the aging immune landscape (Fig. 3), we performed single-cell RNA sequencing (scRNA-seq) of the liver, enriching for CD45⁺ leukocytes along with hepatocytes and endothelial cells (fig. S5A). Uniform manifold approximation and projection (UMAP) clustering (Fig. 4A and fig. S5, B and C) revealed a marked shift in neutrophils from aged WT mice compared with the other three groups (Fig. 4B). Subclustering with pseudotime analysis identified four neutrophil states ordered along a Neu1-to-Neu4 trajectory, with aged WT neutrophils skewed toward Neu2 and Neu3 states (Fig. 4, C and D). Neu2 and Neu3 expanded from near-absence in young and aged cKO mice to ~50 and ~30% of neutrophils, respectively, in aged WT mice (Fig. 4E).

Differential expression showed that Neu1 and Neu4 carried acute immune response and tissue-interaction programs and were unaffected by TRM EP2 deletion. By contrast, Neu2 and Neu3 signatures peaked in aged WT mice and reverted toward youthful expression in aged EP2 cKO mice (Fig. 4F). Neu2 was enriched for proinflammatory alarmins, cyclin-dependent kinase (CDK) inhibitors, and antiapoptotic factors, consistent with senescence-associated secretory (SASP) and cell cycle arrest (CCA) programs, whereas Neu3 showed enrichment for DNA damage response (DDR) genes, proapoptotic factors, and granule protein genes linked to neutrophil extracellular trap (NET) formation. Together, these programs indicate a senescent neutrophil state, incorporating SASP, DDR, apoptosis resistance, and CCA, despite the absence of consensus frameworks for neutrophil senescence and the limited applicability of existing SASP-focused panels to murine immune cells (46–48).

Although neutrophils are considered terminally differentiated, they retain cell cycle machinery (CDK4/6, cyclins, lamin kinases) that is repurposed for nonproliferative functions, including nuclear envelope breakdown during NETosis (CDK4/6), degranulation (CDK5), and control of apoptotic timing (CDK7/9) (49–51). Because existing senescence frameworks may not be designed for neutrophils, in which cell cycle components are coopted for unconventional effector roles, we curated a 57-gene panel detected in >20% of neutrophils to further characterize the senescence and apoptosis programs associated with the Neu2 and Neu3 subclusters. The panel was organized into MICSE-aligned categories (CCA, anti-apoptosis, DDR) with additional modules for pro-apoptosis and NETosis to distinguish apoptotic from senescent states and capture hallmarks of aged neutrophils (fig. S5, D to I). Neu2 scored highest for CCA, SASP, and anti-apoptosis, whereas Neu3 showed the strongest DDR, NETosis, and pro-apoptosis signatures, with aged WT exhibiting the highest scores across categories (Fig. 4G). Composite senescence (CCA + SASP + anti-apoptosis) and stress (DDR + NETosis + pro-apoptosis) scores placed Neu2 in a high-senescence quadrant, with a subset of Neu3 forming a transitional population that progressed toward a high-stress quadrant, consistent with progressive

stress acquisition by senescent neutrophils culminating in apoptosis (Fig. 4H). Thus, Neu2 defines senescent, apoptosis-resistant neutrophils, whereas Neu3 represents stressed neutrophils transitioning toward apoptosis.

Neutrophils age rapidly in circulation, acquiring a CXCR4^{hi}CD62L^{lo} phenotype within 8 to 12 hours (22, 52), making clearance by TRMs in the bone marrow, liver, and spleen essential for immune homeostasis (Fig. 4I). Upon tissue accumulation, aged neutrophils preferentially undergo degranulation and NET formation, amplifying sterile inflammation (53). Flow cytometry of the liver from *Cx3cr1*^{CreER} mice confirmed that aged WT mice accumulated CXCR4⁺ and annexin V⁺ neutrophils despite unchanged total neutrophil numbers, a phenotype reversed in aged cKO mice (Fig. 4, J and K). These data link the transcriptomic Neu2 (senescent) and Neu3 (pro-apoptotic) states to surface phenotype; hereafter, we refer to CXCR4^{hi} neutrophils as senescent neutrophils. Blood neutrophil markers of diapedesis (CXCR2⁺), aging (CXCR4⁺, CD62L⁻), degranulation (CD63⁺), and apoptosis (annexin V⁺) similarly reverted toward youthful amounts in aged EP2 cKO mice (fig. S6, A and B). Liver myeloperoxidase (MPO), a marker of neutrophil burden and NET formation (53), was elevated in aged WT and restored close to youthful amounts in aged EP2 cKO mice (fig. S6C). This phenotype was recapitulated in a second genetic model, the *Rosa26*^{CreER}-inducible global EP2 deletion model with tamoxifen administered at 4 to 6 months of age (fig. S6, D and E). This genetic model overlaps with the *Cx3cr1*^{CreER} model as EP2 is predominantly expressed in macrophages, with minimal basal expression in neutrophils and highest expression in TRMs (fig. S7A).

Liver and plasma abundance of CXCL1 and CXCL2, key CXCR2 ligands driving neutrophil recruitment, were unchanged with aging (Fig. 1, L and M), arguing against increased recruitment as the cause of neutrophil accumulation. We therefore investigated whether accumulation reflected impaired efferocytosis, the process by which professional phagocytes engulf senescent and apoptotic cells. Senescent neutrophils up-regulate phosphatidylserine, annexin A1, and CXCR4 to facilitate macrophage recognition (52, 54), and efferocytosis proceeds through three steps (55): (i) recognition, (ii) ligand-receptor stabilization, and (iii) engulfment (Fig. 4L), followed by lysosomal digestion (56). Different organ TRMs express distinct ligand-receptor pairs across these steps (57).

We examined efferocytosis in young and aged *Rosa26*^{CreER} (WT) and *Rosa26*^{CreER};*ptger2*^{lox/lox} (EP2 global cKO) mice by flow cytometry across liver, spleen, and bone marrow, the three tissues responsible for clearing up to 90% of neutrophils undergoing turnover (27–29) (Fig. 4, M to P, and fig. S7, B and C). In the aged WT liver, F4/80⁺ KCs, the primary efferocytes, were substantially reduced; although total neutrophil numbers were unchanged, CXCR4⁺ senescent and annexin V⁺ apoptotic neutrophils accumulated alongside elevated LFA-1 expression on KCs, consistent with impaired clearance and compensatory stabilization. All changes were largely reversed in aged EP2 cKO mice

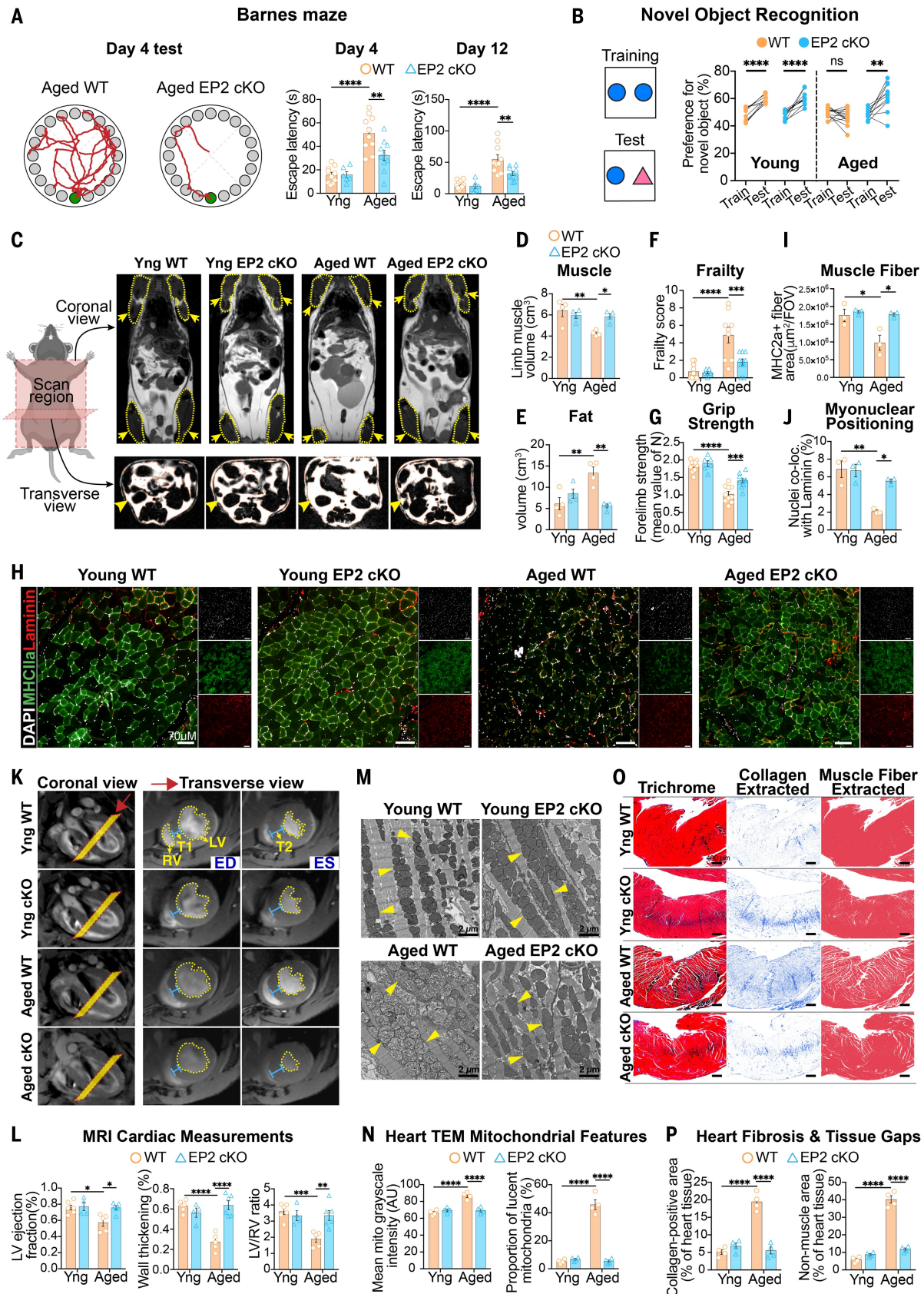


Fig. 2. TRM EP2 deletion prevents age-associated cognitive decline, sarcopenia, adiposity, and decline in cardiac function. Data are mean ± SEM and analyzed using two-way ANOVA with Tukey post hoc tests, unless otherwise stated: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, and **** $P < 0.0001$. Young (yng; 6 to 8 months) and aged (23 to 25 months) C57BL/6J male *Cx3cr1*^{CreER} (WT) and *Cx3cr1*^{CreER}; *EP2*^{lox/lox} (EP2 cKO) mice were examined. (A) Barnes maze assessment of spatial memory. (Left) Representative day-4

test traces after habituation (day 1) and training (days 2 to 3). Escape hole, green; sealed holes, gray. *n* = 6 to 10 mice per group. **(B)** Novel object recognition (NOR) assessment of object-based episodic memory. (Left) Schematic of training and testing phases. (Right) Novel object preference scores across groups. Paired *t* test, training (day 1) versus testing (day 2). *n* = 6 to 10 mice per group. **(C)** MRI of limb muscle and abdominal adipose tissue. (Left) Body positioning and scan regions (pink) in coronal and transverse planes for muscle mass and adipose tissue, respectively. (Right) Representative images from matched anatomical planes, with dashed lines and arrows highlighting limb muscle (top) and adipose tissue (bottom). **(D and E)** MRI quantification of limb muscle volume (D) and adipose tissue volume (E). *n* = 4 mice per group. **(F)** Frailty assessment using the mouse frailty index (see methods), which assesses eight categories scored as 0 = absent, 0.5 = mild, or 1 = severe. *n* = 6 to 10 mice per group. **(G)** Grip strength measurements averaged over 10 trials per mouse; force is reported in Newtons (N). *n* = 6 to 10 mice per group. **(H)** Immunofluorescence of calf muscle cross sections. Large panels show merged channels; small panels show individual 4',6-diamidino-2-phenylindole (DAPI) (nuclei, white), MHC IIa (muscle fibers, green), and Laminin (basal lamina, red). **(I and J)** Quantification of muscle fiber area (I) and nucleus localization (J); each data point represents the average of three fields of view (FOVs) per animal. *n* = 3 mice per group. **(K)** Cardiac MRI in coronal (left) and transverse views at end-diastole (ED; middle) and end-systole (ES; right). Yellow plane and red arrow in coronal view indicate transverse imaging position at the widest ventricular width. Yellow dashed lines outline LV and RV; blue bars indicate septal wall thickness (T) at ED (T1) and ES (T2). **(L)** Quantification of LV ejection fraction (EF), septal wall thickness, and LV/RV ratio from (K): $LVEF = (ED\ volume - ES\ volume) / ED\ volume$; wall thickening = $(T2 - T1) / T1$; LV/RV volume ratio at ED phase. *n* = 4 to 6 mice per group. **(M)** Representative TEM images of cardiomyocytes in LV heart muscle, positioned at comparable regions across subjects, acquired at 4000 $\times$ magnification. Arrows indicate mitochondria. The TEM images shown in this panel are also presented in fig. S4K (first column), where they are displayed alongside additional processing and magnifications used for related quantifications. **(N)** Quantification of mitochondrial features from heart TEM. Mean mitochondrial grayscale intensity (AU, arbitrary units) reflects the brightness of extracted mitochondria on a scale of 0 (dark) to 255 (white). The proportion of abnormal lucent mitochondria (percent) is measured from images taken at 1000 $\times$ magnification, with lucency defined as grayscale intensity >80 AU. Each data point represents the average of four FOVs per animal. *n* = 4 mice per group. **(O)** Representative trichrome staining of heart tissue to assess collagen deposition and cardiomyocyte structure. Sections were taken along the longitudinal axis of the LV from a similar region across subjects. Images were acquired at 2.5 $\times$ magnification. Image deconvolution of trichrome stain (left column) identifies collagen (blue; middle column) and muscle (red; right column). **(P)** Quantification of heart tissue composition from trichrome staining in (O). Collagen-positive area is quantified from collagen-extracted images, and nonmuscle area from muscle fiber-extracted images. Each datapoint represents the average from three FOVs per animal. *n* = 4 mice per group.

(Fig. 4, M and N). In the spleen, where EP2 was highly expressed in TRMs but minimal in neutrophils (fig. S7B), aged WT mice showed increased total neutrophils and accumulation of CXCR4⁺ senescent and annexin V⁺ apoptotic neutrophils alongside elevated LFA-1 on splenic TRMs, all restored in aged EP2 cKO mice (Fig. 4O). Bone marrow showed the same pattern: CXCR4⁺ senescent neutrophil accumulation and elevated LFA-1 on TRMs, both reversed in aged EP2 cKO mice (Fig. 4P). As in the liver and spleen, EP2 expression in the bone marrow was largely restricted to macrophages (fig. S7C). Mer tyrosine kinase (MerTK), which mediates engulfment of phosphatidylserine-exposing cells by bone marrow macrophages (58), was elevated in aged WT TRMs and restored in aged cKO mice. Across all clearance organs, elevated integrin expression accompanied senescent neutrophil accumulation, suggesting a compensatory response to chronic efferocytic failure. Together, aging drives senescent neutrophil accumulation across all major efferocytic organs alongside compensatory up-regulation of stabilization receptors, both of which are prevented by TRM EP2 deletion.

TRM-specific EP2 deletion restores efferocytosis and limits senescent neutrophil-driven tissue injury

To directly test whether TRM EP2 deletion restores efferocytic capacity, we examined liver KCs and splenic red pulp macrophages (RPMs) from young and aged *Cx3cr1*^{CreER} and *Cx3cr1*^{CreER};EP2^{lox/lox} mice, in which EP2 deletion is largely TRM-restricted (Fig. 1G). TRMs were coincubated ex vivo with carboxyfluorescein succinimidyl ester (CFSE)-labeled target cells, with 4°C incubation to distinguish adhesion from uptake (fig. S8). KCs were challenged with senescent neutrophils or dexamethasone-induced apoptotic thymocytes (59), whereas RPMs were challenged with apoptotic RBCs (60) or lymphocytes (61), with >95% labeling efficiency across substrates (fig. S8).

Aged WT TRMs from both organs showed considerably reduced efferocytosis of all target cells, restored by TRM EP2 deletion (Fig. 5A). The defect was most pronounced for senescent neutrophils (~10-fold reduction versus young, ~sixfold reduction versus cKO), compared with ~2.5-fold reduction for apoptotic substrates. Because senescent and apoptotic cells share phosphatidylserine-dependent recognition pathways (TIM4 and MerTK) (62), age-associated defects in recognition likely contribute broadly to the clearance deficit, consistent with reported MerTK cleavage in aging (63). However, the selective severity for senescent neutrophils suggests an additional failure at the stabilization step, supported by compensatory up-regulation of the integrin LFA-1 on aged WT TRMs across efferocytic organs (Fig. 4, M to P).

To examine age-associated efferocytic deficits in situ, we used MACSima multiplexed immunohistochemistry (mIHC) (64) to map KCs, neutrophils, and degranulation markers in liver from young and aged *Cx3cr1*^{CreER} and EP2 cKO mice. KCs were identified by CD169 (Siglec-1), a sialoadhesin marking TRMs with high specificity (65), and neutrophils were stained with Ly6G, the senescence marker CXCR4, and myeloperoxidase (MPO), a neutrophil granule enzyme elevated during degranulation and NETosis that promotes oxidative tissue injury (66) (Fig. 5B). The aged WT liver showed reduced CD169⁺ KC area alongside increased CXCR4⁺ and MPO⁺ neutrophil burden; all three measures were restored toward youthful states in aged EP2 cKO mice (Fig. 5, B and C). Spatial colocalization of CD169⁺ KCs with Ly6G⁺ neutrophils, a proxy for efferocytic engagement, was reduced in aged WT but restored in aged EP2 cKO mice, as was the elevated CXCR4⁺Ly6G⁺ neutrophil pool (Fig. 5, B and C). The increase in MPO is consistent with enhanced NET-associated sterile inflammation (66). Consistent with the periportal enrichment of KCs (67), conventional immunohistochemistry (IHC) using the macrophage marker F4/80 confirmed portal localization across groups with no zonal redistribution, whereas Ly6G⁺ neutrophils were distributed across both portal and central hepatic zones without zonal preference (fig. S9, A and B). Larger-area conventional IHC similarly showed elevated MPO and elastase in aged WT liver, restored in aged EP2 cKO mice (fig. S9C).

Parallel conventional IHC of spleen showed reduced colocalization of Ly6G⁺ neutrophils with the lysosomal phagocyte marker CD68, indicating impaired degradation of senescent neutrophils. CD68⁺ macrophage area was also diminished in the aged WT spleen, with both defects restored in aged EP2 cKO mice. Aged WT spleens accumulated total and CXCR4⁺ senescent neutrophils and showed increased MPO and elastase staining indicating neutrophil-driven tissue injury, all prevented in aged EP2 cKO mice (fig. S9, D and E). These findings demonstrate that impaired clearance in aging extends across efferocytic organs and is enabled by TRM EP2 signaling.

Senescent cells can exert deleterious effects on neighboring cells through SASP-mediated paracrine signaling (68). To investigate paracrine consequences of senescent neutrophil accumulation, we used mIHC to examine hepatic parenchymal cells. In the aged WT liver, hepatocytes (E-cadherin⁺) up-regulated CD54 (ICAM-1), an adhesion molecule induced on hepatocytes during inflammatory stress (69); sinusoidal endothelial cells (CD31⁺) showed elevated CD106 (VCAM-1), a marker of endothelial activation (70); and stellate cells (CD206⁺), the principal source of hepatic fibronectin (FN) (71), increased FN deposition (Fig. 5D).

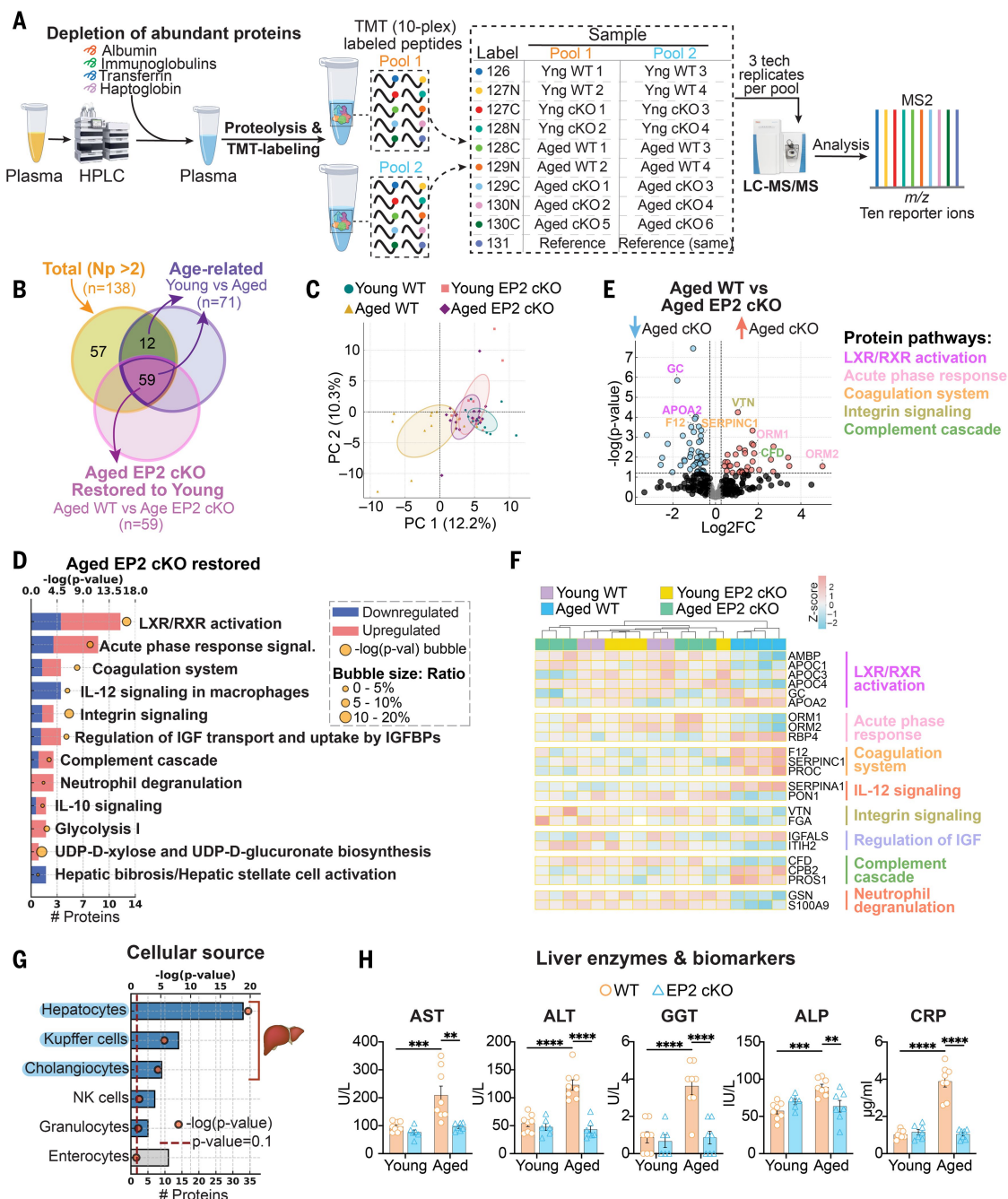


Fig. 3. TRM EP2 deletion prevents age-associated changes in the plasma proteome. Data are mean $\pm$ SEM and analyzed using two-way ANOVA with Tukey post hoc tests, unless otherwise stated: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, and **** $P < 0.0001$. Young (yng; 6 to 8 months) and aged (23 to 25 months) C57BL/6J male *Cx3cr1^{CreER}* (WT) and *Cx3cr1^{CreER};EP2^{lox/lox}* (EP2 cKO) mice were examined. **(A)** Plasma TMT-based shotgun proteomics workflow. After depletion of albumin, immunoglobulins, and transferrin, and subsequent protein digestion, peptides were labeled with 10 isobaric TMT tags (m/z 126 to 131), with label assignments color coded for visualization, corresponding to reporter ion positions on the MS2 m/z spectrum used for protein quantification. **(B)** Venn diagram showing the overlap among protein groups for the 138 proteins detected with number of peptides (Np) > 2. $n = 4$ to 6 mice per group. **(C)** PCA of plasma proteomes for 138 proteins. Each point represents a technical replicate from one biological sample. $n = 4$ to 6 mice per group, with three technical replicates per sample. **(D)** Pathway enrichment analysis of plasma proteins that were restored to youthful amounts in aged EP2 cKO mice. Bar length indicates the number of differentially expressed proteins per pathway, with red and blue denoting up- and down-regulated proteins, respectively. Bubble position along the top axis indicates pathway significance [$-\log(P\text{-value})$]. Bubble size represents the ratio of detected proteins in this dataset to the total number of proteins associated with that pathway in Ingenuity Pathway Analysis (IPA). **(E)** Differentially expressed plasma proteins between aged WT and aged EP2 cKO mice. Red and blue dots represent proteins meeting significance thresholds (FDR < 0.05, foldchange > 1.23). Selected annotated proteins are color coded by protein pathway (right). **(F)** Heatmap of representative plasma proteins restored to youthful amounts with TRM EP2 cKO. Proteins are grouped by pathway and color coded by functional category on the right. Values represent the average of three technical replicates per sample. $n = 4$ to 6 mice per group. **(G)** Cell type enrichment of age-associated plasma proteins restored to youthful amounts with TRM EP2 cKO. Bar length indicates the number of mapped proteins; bubble position reflects $-\log(P\text{-value})$; the dashed red line marks the $P\text{-value} = 0.1$ threshold; blue bars indicate significant enrichment, gray indicates nonsignificant. **(H)** Liver enzymes and biomarkers measured in serum. AST, aspartate aminotransferase; ALT, alanine aminotransferase; GGT, gamma-glutamyltransferase; ALP, alkaline phosphatase; CRP, C-reactive protein. $n = 6$ to 8 mice per group.

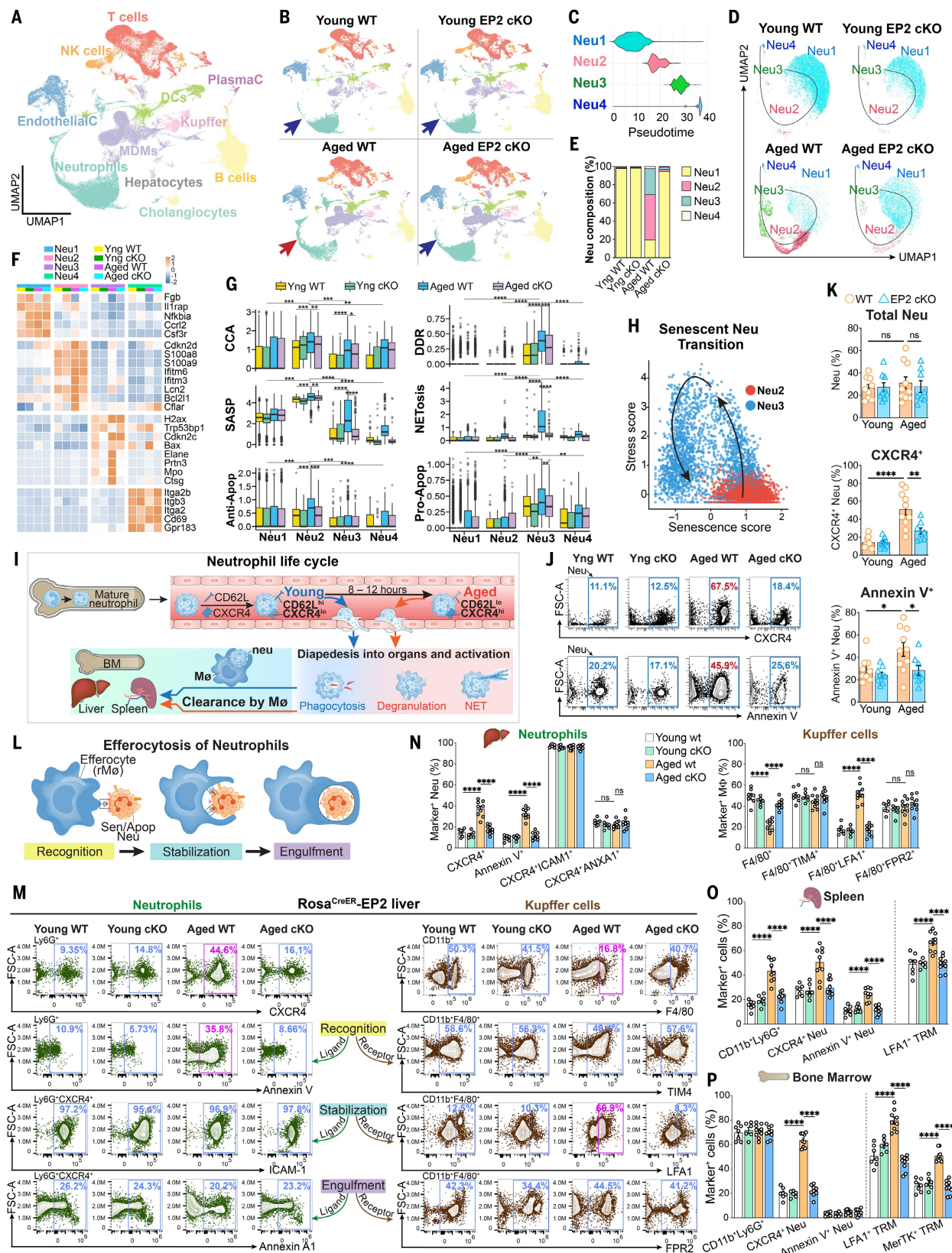


Fig. 4. TRM EP2 deletion reduces senescent neutrophil burden and restores efferyocytic machinery across aged efferyocytic organs. Young (yng; 6 to 8 months) and aged (23 to 25 months) male *Cx3cr1*^{CreER} (WT) and *Cx3cr1*^{CreER}; *EP2*^{lox/lox} (EP2 cKO) mice were examined (A to K). (M to P) show data from *Rosa26*^{CreER} (WT) and *Rosa26*^{CreER}; *EP2*^{lox/lox} (EP2 cKO) mice of the same age groups. Data were analyzed using two-way ANOVA with Tukey post hoc tests, except for (G) (Wilcoxon rank-sum test); * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$. (A) UMAP visualization of liver scRNA-seq integrating all four groups. Major cell populations are annotated using color coded labels matching their corresponding clusters. DCs, dendritic cells; plasmaC, plasma cells; endothelialC, endothelial cells; Kupffer cells, KCs. $n = 8$ mice total. (B) UMAP visualization of liver

scRNA-seq by experimental group. Arrows highlight the neutrophil cluster. $n = 2$ mice per group. (C) Pseudotime trajectory of Neu (neutrophils) integrating all four experimental groups. (D) UMAP visualization of neutrophil subclusters (Neu1 to 4) split by experimental groups, showing condition-dependent shifts in subcluster distribution. (E) Mean percentage of each neutrophil subcluster (Neu1 to 4) within total neutrophils across experimental groups. (F) Heatmap of top differentially expressed genes across neutrophil subclusters (Neu1 to 4) and age/genotype conditions. Scale bar indicates z-scored expression. (G) Module scores for CCA, SASP, anti-apoptosis, DDR, NETosis, and pro-apoptosis gene categories across neutrophil subclusters (Neu1 to 4) and experimental groups. Each data point represents a single cell. The y-axis represents per-cell mean expression of genes in each category, adjusted for background expression. Box plots show median (thick line), interquartile range (box), and whiskers ($1.5 \times$ IQR). Significance lines compare subclusters (long) or experimental groups within a subcluster (short). CCA, cell cycle arrest; SASP, senescence-associated secretory phenotype; Apop, apoptosis; DDR, DNA damage response. (H) Composite senescence (CCA + SASP + anti-apoptosis) versus stress (DDR + NETosis + pro-apoptosis) scores for Neu2 and Neu3 cells. Arrow indicates the cell state transition trajectory. Each data point represents a single cell. (I) Schematic of the neutrophil life cycle illustrating stages from bone marrow maturation to circulation, diapedesis into tissues, activation, and clearance by macrophages. Blue and red arrows indicate the trafficking and fate of young and aged neutrophils, respectively. (J) Representative flow cytometry plots of CXCR4 (top) and annexin V (bottom) expression within liver neutrophils (Neu) across experimental groups. $n = 8$ to 10 mice per group. (K) Quantification of total neutrophil percentage within CD11b⁺ cells, and CXCR4⁺ and annexin V⁺ percentages within neutrophils, from (J). $n = 8$ to 10 per group. (L) Schematic of the three steps of neutrophil (Neu) efferocytosis by resident macrophages (rMΦ): recognition, stabilization, and engulfment. Dashed boxes highlight ligand–receptor interactions at each step, such as efferocyte TIM4–neutrophil PtdSer (recognition), efferocyte integrin–neutrophil ICAM-1/VCAM-1 (stabilization), and efferocyte FPR2–neutrophil annexin A1 (engulfment). Sen, senescent; Apop, apoptotic. (M) Flow cytometry strategy with representative plots of liver neutrophils and KCs, showing cell abundance, neutrophil apoptosis and senescence, and expression of ligand–receptor pairs involved in senescent neutrophil clearance, corresponding to efferocytosis stages outlined in (L). The parent population for each marker is indicated in the upper left of each plot row. (N) Quantification of neutrophil (left) and KC (right) marker percentages from (M). Neutrophil markers are within Ly6G⁺ cells; KC markers are within CD11b⁺ cells. $n = 6$ to 9 per group. (O) to (P) Quantification of neutrophil and TRM marker percentages in spleen (O) and bone marrow (P). Experimental group annotations (yng WT, yng cKO, aged WT, aged cKO) follow the color legend shown in (N). $n = 6$ to 9 mice per group.

Colocalization of E-cadherin with CD54, CD31 with CD106, and CD206 with FN was substantially increased with age but restored to youthful amounts in aged EP2 cKO mice (Fig. 5, D and E). Spatial analysis revealed reduced distances between stressed parenchymal cells and Ly6G⁺CXCR4⁺ senescent neutrophils in aged WT mice, which was normalized in aged EP2 cKO mice, supporting proximity-dependent paracrine signaling. Collectively, EP2-mediated impairment of TRM efferocytosis in aging promotes senescent neutrophil accumulation, NET-mediated tissue injury, and paracrine inflammatory and fibrotic stress, all of which are prevented by TRM EP2 deletion.

TRM EP2 deletion restores integrin activation and efferocytic programs in aging

Given the impaired efferocytosis observed across liver, spleen, and bone marrow (Figs. 4 and 5), we interrogated TRM–neutrophil interactions at single-cell resolution. CellChat analysis (72) of liver scRNA-seq from young and aged WT and EP2 cKO mice identified integrins as the dominant receptor class in KCs, with cognate ligands (e.g., *Icam1*, *Thbs1*) enriched in neutrophils, forming extensive ligand–receptor networks (Fig. 6A and fig. S10, A and B). Flow cytometry confirmed elevated LFA-1 (CD11a/CD18; Itgal/Itgb2) on aged TRMs across tissues (Fig. 4), implicating dysregulation of the $\beta 2$ integrin subunit (Itgb2), which is shared by all major leukocyte integrin heterodimers (73), as a common feature of age-associated efferocytic failure.

We therefore investigated how $\beta 2$ integrins are activated during efferocytosis. As illustrated in Fig. 6B, activation proceeds through four stages: (i) resting (bent, inactive); (ii) ligand binding, where Rac1 and RhoA engage the $\beta 2$ tail to initiate cytoskeletal remodeling (74); (iii) membrane protrusion, where Skap1 facilitates membrane extension and transitions integrins to an extended-closed state; and (iv) stabilization, where Tln2 and Fermt3 induce the fully extended, high-affinity conformation enabling stable neutrophil engagement (74, 75). Although most integrin subunits were reduced in aged KCs, *Itgb2* was increased, consistent with compensatory up-regulation in the setting of impaired activation. By contrast, all five key coactivators (*Rac1*, *RhoA*, *Skap1*, *Tln2*, *Fermt3*) were reduced with aging but restored in aged EP2 cKO mice (Fig. 6C), with pathway analysis confirming suppression of integrin activation in aged WT KCs (Fig. 6D).

Because integrin stabilization is required for engulfment and downstream clearance, we next examined post-engulfment programs (fig. S10C): Immediate early genes (*Jun*, *Egr1*) initiate the transcriptional response; post-engulfment regulators (*Arg1*, *Il1rn*) coordinate anti-inflammatory signaling; nuclear receptors (*Lxr*, *Pparg*) mediate lipid degradation; and efflux transporters (*Abca1*, *Abcg1*) export lipid

contents (76–78). All of these effectors were dysregulated in aged WT KCs, with activators down-regulated and suppressors elevated, a pattern reversed in aged EP2 cKO mice (fig. S10D). Pathway analysis confirmed broad suppression of phagocytic programs in aged WT KCs (Fig. 6E). Among 1227 age-dysregulated genes rescued with EP2 deletion (fig. S10E), efferocytosis-related pathways, including integrin signaling, liver X receptor (LXR)/RXR activation, and coordinated lysosomal expression and regulation (CLEAR) network signaling, were among the most restored programs (fig. S10F). Additional age-associated disruptions of immune regulation, homeostasis, and stress adaptation were also normalized (fig. S10G), supporting the metabolic demands of efficient clearance.

Upstream transcription factor (TF) analysis revealed coordinated regulation of efferocytosis-associated TFs (Fig. 6F). Aging was associated with suppression of *JUNB*, *STAT3*, and *RELA*, TFs that promote phagocytic clearance (79–85), and induction of CREB1 that impairs efferocytosis (81, 85), a pattern reversed in aged EP2 cKO mice (Fig. 6G). Together, these data demonstrate that EP2 signaling disrupts efferocytosis in aging TRMs by impairing transcriptional control, integrin activation, and post-engulfment clearance, defects that are restored by TRM-specific EP2 deletion.

Pharmacologic EP2 antagonism restores neutrophil clearance in aged mice

To validate our genetic findings, we treated 22-month-old male mice with the selective EP2 antagonist PF-04418948 (PF) (86) for 2 months (Fig. 6H). PF normalized circulating total, CXCR4⁺ senescent, and CD62L[−] aged neutrophils toward youthful amounts (Fig. 6I and fig. S11A). In the liver, PF similarly reduced total and CXCR4⁺ neutrophils, partially restored KC abundance, and decreased the proportion of LFA-1⁺ KCs, suggesting a partial reversal of the age-associated integrin-stabilization defect (Fig. 6J and fig. S11, B and C). Other efferocytic ligand–receptor pairs on neutrophils and KCs were unchanged (fig. S11C), reinforcing $\beta 2$ integrin dysregulation as the dominant EP2-dependent defect.

PF treatment also reduced age-associated senescent neutrophil accumulation in spleen and bone marrow. In spleen, total Ly6G⁺ neutrophils and annexin V⁺ apoptotic neutrophils were elevated with aging and substantially reduced by PF, with a trend toward normalization of CXCR4⁺ neutrophils (Fig. 6K and fig. S11, D and F). In bone marrow, PF normalized CXCR4⁺ senescent neutrophils and rescued age-related loss of VCAM-1⁺ neutrophils, suggesting that EP2 inhibition reestablishes normal neutrophil retention in the bone marrow niche (Fig. 6L and fig. S11, E and G). LFA-1⁺ TRM proportions outside the liver were not substantially restored, indicating that integrin normalization may

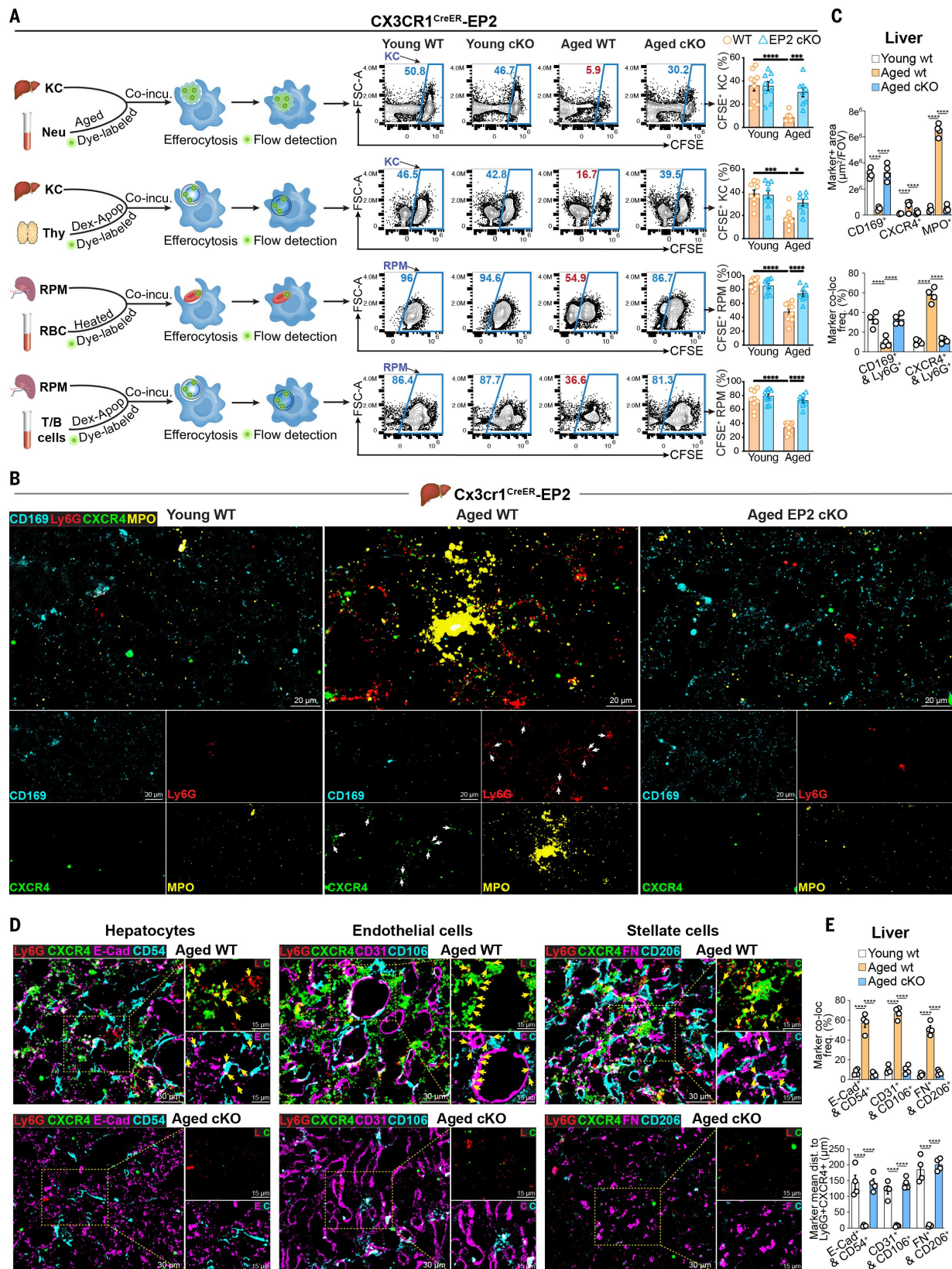


Fig. 5. TRM EP2 deletion restores clearance of senescent and apoptotic cells and reduces senescent neutrophil-driven damage and paracrine stress in aged tissues. Young (yng; 6 to 8 months) and aged (23 to 25 months) male *Cx3cr1^{CreER}* (WT) and *Cx3cr1^{CreER};EP2^{lox/lox}* (EP2 cKO) mice were examined. Data are mean $\pm$ SEM. (A) was analyzed using two-way ANOVA with Tukey post hoc tests; (C and E) were analyzed using one-way ANOVA with Tukey post hoc tests: * $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, **** $P < 0.0001$. For (B) to (E), colocalization (co-loc) was defined as overlapping signals or proximity of $<1 \mu\text{m}$ between detected signal centers, and images across groups were acquired from comparable anatomical regions using identical settings for each stain panel. (A) Ex vivo efferocytosis assays. CFSE-labeled target cells were co-incubated

(co-incu.) with liver or spleen cell preparations; CFSE uptake was gated on TRMs (liver KCs; splenic RPMs). (Left) Experimental schematics; (middle); representative flow plots. (Right) Quantification. Target cells: aged blood neutrophils (Neu), dexamethasone-induced apoptotic (Dex-Apop) thymocytes (Thy), heat-treated apoptotic RBCs, or Dex-Apop T/B cells. Representative plots for KC–Neu and KC–Thy assays (first two rows) are the 37°C condition from the full control series shown in fig. S8A, which includes matched w/o target cell and 4°C negative controls. Analogous control series for the RPM–RBC and RPM–T/B assays are shown separately in fig. S8B. Kupffer cells, KCs; RPMs, red pulp macrophages. $n = 8$ to 10 mice per group. (B) Representative MACSima multiplexed immunohistochemistry (mIHC) of liver sections stained for CD169 (KCs), Ly6G (neutrophils), CXCR4 (senescence), and MPO (degranulation/NETosis). (Top row) merged channels; (middle and bottom rows) each individual channel of the merged image above, displayed separately. Arrows indicate Ly6G and CXCR4 colocalization in aged WT. Scale bars, 20 μm . (C) Quantification of imaging data from (B): marker-positive area (CD169, CXCR4, MPO) and colocalization frequency (CD169&Ly6G, CXCR4&Ly6G). Each data point represents the average of three FOVs per animal. FOV, field of view. $n = 4$ mice per group. (D) Representative MACSima mIHC of liver sections showing paracrine stress markers in three parenchymal cell populations: hepatocytes (E-Cadherin/CD54), sinusoidal endothelial cells (CD31/CD106), and stellate cells (FN/CD206), each costained with Ly6G and CXCR4 to mark senescent neutrophils. Dashed boxes indicate magnified insets; top insets show Ly6G (L) and CXCR4 (C) channels; bottom insets show individual parenchymal marker pairs, labeled by first letter and color-matched to their respective channels in the merged image. Yellow arrows highlight proximity of stressed parenchymal cells to CXCR4⁺ neutrophils. Scale bars, 30 μm (main) and 15 μm (insets). (E) Quantification of paracrine stress from (D). (Top) Colocalization frequency of parenchymal cell–stress marker pairs (E-Cad&CD54, CD31&CD106, FN&CD206). (Bottom) Mean spatial distance of each stressed parenchymal marker pair to Ly6G⁺CXCR4⁺ senescent neutrophils. Each data point represents the average of three FOVs per animal. $n = 4$ mice per group.

require longer or earlier treatment (Fig. 6, K and L). Other efferocytic ligand-receptor pairs on neutrophils and TRMs in spleen and bone marrow were similarly unchanged (fig. S11, F and G).

To directly assess neutrophil efferocytosis by TRMs, we performed ex vivo coculture assays using senescent blood neutrophils and freshly enriched liver KCs, as in Fig. 5A (Fig. 6M). KCs from PF-treated aged mice showed considerably increased neutrophil uptake compared with vehicle controls, restoring efferocytic capacity to near-youthful states (Fig. 6, M and N, and fig. S11H). Together, these results demonstrate that pharmacologic EP2 inhibition partially reverses age-associated TRM dysfunction and senescent neutrophil accumulation, with strongest rescue in the liver.

TRM-neutrophil dysregulation in human aging and disease

We next investigated whether similar TRM-neutrophil alterations occur in human aging and disease. Analysis of scRNA-seq data from the Williams Human Liver Cell Atlas (67) revealed that aged healthy livers showed reduced TRM proportions and expanded neutrophil populations relative to younger subjects, with MDMs largely unchanged (Fig. 6, O and P). A similar pattern, with loss of TRMs and neutrophil expansion, was observed in diseased livers (Fig. 6P). PTGER2 expression was selectively increased in TRMs from both aged and diseased human livers, with smaller increases in neutrophils and MDMs (Fig. 6Q and fig. S12A).

Human liver neutrophils also exhibited age- and disease-associated shifts toward senescent-like states. We used orthologous gene sets for SASP, anti-apoptosis, and DDR gene sets, three hallmark categories with sufficient expression in the scRNA-seq data to assess senescence features (fig. S12, B to D). In the aged liver, neutrophils transitioned from Neu1+2 clusters toward Neu3+4 clusters (Fig. 6R). Neu3+4 neutrophils displayed considerably higher SASP ($P = 4.7 \times 10^{-7}$) and anti-apoptosis ($P = 7.2 \times 10^{-5}$) module scores compared with Neu1+2 neutrophils, consistent with a senescent-like phenotype (Fig. 6S). CellChat analysis revealed reduced TRM-neutrophil efferocytic interactions in the aged liver, with losses in integrin, galectin-9, annexin A1, and NAMPT signaling that were accompanied by a gain in MHC-II antigen presentation pathways (Fig. 6T and table S1). These changes suggested a shift from tolerogenic neutrophil clearance to immunostimulatory signaling, in line with precedent that defective efferocytosis enhances macrophage MHC-II signaling and CD4⁺ T cell activation in age-associated inflammation (87).

Similar changes were observed in the diseased liver, where neutrophils showed a more pronounced senescent transcriptional signature, shifting from Neu1+5 toward Neu2+3+4 populations (Fig. 6U), with elevated SASP ($P = 5.7 \times 10^{-78}$), anti-apoptosis ($P = 9.8 \times 10^{-48}$), and DDR ($P = 1.5 \times 10^{-5}$) scores (Fig. 6V). TRM-neutrophil communication showed comparable reductions in interaction number and strength (Fig. 6W and table S1). Analysis of a second organ, the heart, using the

Tabula Sapiens heart atlas (88) also revealed age-associated loss of TRMs and selective up-regulation of *PTGER2* in TRMs, however without expansion of neutrophils, consistent with the heart not being a primary efferocytic site for neutrophils (fig. S12, E to G). These cross-organ findings in humans, although correlative, are consistent with the relationships observed in our mouse models. Future studies in human samples will be needed to directly test whether aged TRMs show reduced efferocytic capacity against senescent neutrophils and whether EP2 antagonism can restore this function.

As a complementary genetic approach, we examined whether EP2 or PGE₂ pathway genes are associated with human longevity in published genome-wide association studies (GWAS). Across six GWAS studies of longevity and healthspan (89–93), *PTGER2* and related PGE₂ pathway genes showed no significant associations (fig. S12H). A Manhattan plot of the Timmers 2020 healthspan GWAS revealed no significant loci on chromosome 14 where *PTGER2* resides (fig. S12I). The top longevity loci mapped to *APOE*, *LPA*, *CDKN2B-AS1*, *ATXN2*, and related genes within pathways including lipid metabolism, neurodegeneration, cell cycle and senescence, cardiovascular function, immune function, DNA repair, insulin/IGF signaling, and proteostasis (fig. S12, I and J). Notably, several of these pathways, particularly cell cycle and senescence and immune function, are directly relevant to the TRM-neutrophil clearance mechanism identified in our study, suggesting that existing GWAS may not yet capture the full spectrum of genes contributing to these biological processes. Consistent with this, epidemiological studies have widely reported associations between *PTGER2* and aging-related diseases including Alzheimer's disease, immune disorders, and cancer (94).

Discussion

We identify TRMs as central regulators of multiorgan aging through PGE₂-EP2 signaling. EP2 deficiency in aging TRMs restored mitochondrial fitness and reduced systemic and organ-level phenotypes of aging, including inflammation, cognitive decline, sarcopenia, frailty, adiposity, and cardiac fibrosis. Mechanistically, TRMs drive aging through EP2-dependent suppression of efferocytosis of senescent neutrophils. In aging TRMs, EP2 signaling disrupts transcriptional programs required for integrin activation and phagocytic clearance, leading to senescent neutrophil accumulation, persistent degranulation, NETosis, and propagation of sterile inflammation. EP2 inhibition preserves TRM efferocytic capacity and mitigates neutrophil-driven tissue aging.

Senescent neutrophils, defined here by features including CCA, SASP, and apoptosis resistance, represent an understudied state of neutrophil dysfunction in aging. Classical senescence frameworks have focused on proliferative cells (46, 48); however, neutrophils retain cyclin-dependent kinases (CDK4/6, CDK5, and CDK7/9) that are repurposed for NETosis, degranulation, and apoptotic timing (49–51). In aged neutrophils, increased expression of CDK inhibitors appears to constrain

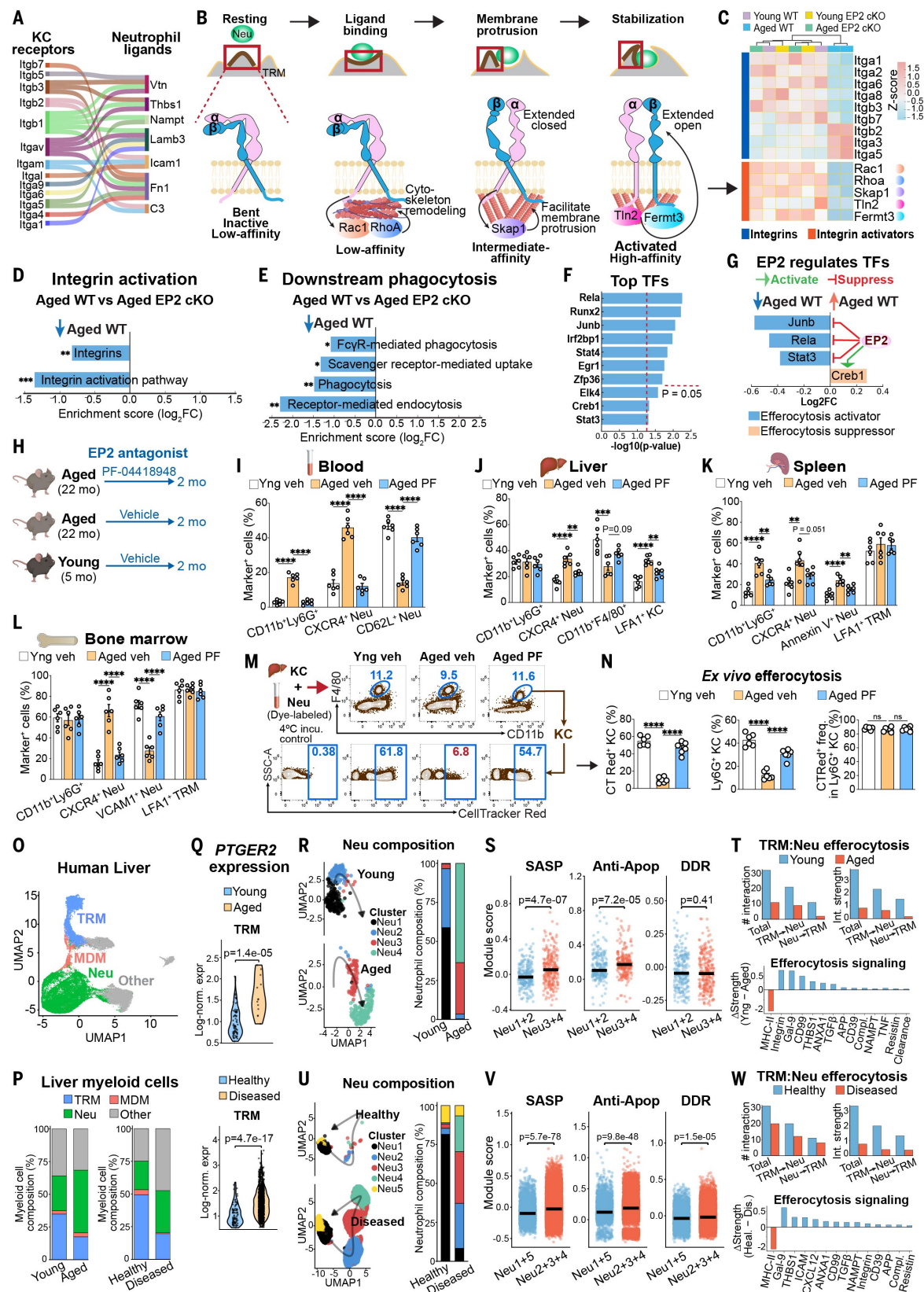


Fig. 6. EP2 impairs efferyocytic stabilization and post-engulfment programs in aged KCs and pharmacological EP2 inhibition restores neutrophil clearance in aged mice and TRM-neutrophil alterations occur in aged and diseased human organs. (A to G) scRNA-seq from young (6 to 8 months) and aged (23 to 25 months) male *Cx3cr1*^{CreER} (WT) and *Cx3cr1*^{CreER};EP2^{lox/lox} (EP2 cKO) mice. (H to N) young (yng; 7 months) and aged (24 months) male C57BL/6J mice treated with vehicle (veh) or PF-04418948 (PF). (O to W) human liver scRNA-seq data (Guilliams *et al.*, PMID: 35021063). Data are mean ± SEM (I) to (N) were analyzed using one-way ANOVA with Tukey post hoc tests; (O) to (W) were analyzed using Wilcoxon rank-sum test: *P < 0.05, **P < 0.01, ***P < 0.001, ****P < 0.0001. (A) Ligand–integrin mapping between

neutrophils and KCs using CellChat, filtered for biologically relevant expression and communication strength. (B) Schematic of integrin activation stages during efferocytosis. Integrins function as α/β heterodimers transitioning from a low-affinity (bent) to high-affinity (extended, open) state. Coactivators *Rac1*, *RhoA*, *Skap1*, *Tln2*, and *Fermt3* bind the cytoplasmic tail of β -subunits to promote cytoskeletal remodeling, membrane protrusion, and integrin unbending. Neu, neutrophils. (C) Expression of integrins and integrin-activating genes in KCs. Expression values represent per-sample averages across single cells. Colored dots next to each integrin activator correspond to the color used for that factor in the schematic in (B). (D) Enrichment analysis of integrin and integrin-activating pathways in KCs. Negative values indicate decreased enrichment in aged WT compared with aged EP2 cKO mice. $**P < 0.01$, $***P < 0.001$. (E) Enrichment analysis of KC downstream phagocytic pathways regulated by TRM EP2. Pathway enrichment was calculated based on genes associated with post stabilization phases of efferocytosis. Negative enrichment scores indicate reduced pathway activity in aged WT compared with aged EP2 cKO. $*P < 0.05$; $**P < 0.01$. (F) Top predicted TFs regulating integrin- and efferocytosis-related phagocytic program genes. Genes from (C) to (E) were jointly analyzed. Significance is shown as $-\log_{10}(P\text{-value})$; red dashed line indicates $P = 0.05$. (G) EP2-regulated TFs differentially expressed between aged WT and aged EP2 cKO KCs. Bar direction reflects expression change; leftward bars indicate reduced TF expression in aged WT and restoration in aged EP2 cKO. Bar length represents $\log_2$ fold change. Bar color indicates each TF's reported role in efferocytosis: blue for activators, orange for suppressors. (H) Schematic of pharmacological EP2 inhibition through oral gavage in young (5 months) and aged (22 months) male C57BL/6J mice. (I) Quantification of blood neutrophil ($CD11b^+Ly6G^+$) marker percentages across treatment groups. $n = 6$ mice per group. (J) Quantification of liver neutrophil ($CD11b^+Ly6G^+$) and KC ($CD11b^+F4/80^+$) marker percentages across treatment groups. $n = 6$ mice per group. (K) to (L) Quantification of neutrophil and TRM marker percentages in the spleen (K) and bone marrow (L). $n = 6$ mice per group. (M) Ex vivo efferocytosis assay. CellTracker red-labeled aged blood neutrophils were cocultured with liver cell preparations; uptake was gated on KCs ($CD11b^+F4/80^+$). Representative flow plots of CellTracker Red⁺ KCs are shown, with 4°C incubation controls. (N) Quantification of CellTracker Red⁺ KCs (left), $Ly6G^+$ KCs (middle), and CellTracker Red⁺ frequency within $Ly6G^+$ KCs (right) from (M). Representative flow plots for $Ly6G^+$ KCs and CellTracker Red⁺ $Ly6G^+$ KCs are shown in fig. S11H. $n = 6$ mice per group. (O) UMAP of human liver myeloid cells from human scRNA-seq data (Guilliams *et al.*, PMID: 35021063) integrating all donors. This dataset is used for (O) to (W). (P) Myeloid cell composition in human liver. (Left) Young (ages 49, 53; $n = 2$ donors) versus aged (ages 73 to 75; $n = 2$ donors) in healthy livers (cholecystolithiasis). (Right) Healthy (cholecystolithiasis; $n = 4$ donors) versus diseased ($n = 10$ donors; hepatocellular carcinoma, hepatocellular adenoma, colorectal cancer with liver metastases, liver injury, morbid obesity). The same donor groupings apply to (Q) to (W). (Q) *PTGER2* expression in human liver TRMs. (Top) Young versus aged (healthy donors). (Bottom) Healthy versus diseased. Each data point represents a single cell expressing *PTGER2*. (R) Neutrophil subclustering in healthy donors. (Left) UMAP of neutrophil subclusters (Neu1 to 4) split by young and aged, with pseudotime trajectory (arrows). (Right) Neutrophil subcluster composition by age group. (S) Senescence module scores (SASP, anti-apoptosis, DDR) comparing Neu1+2 (young-enriched) versus Neu3+4 (aged-enriched) subclusters in healthy donors. Each data point represents a single cell. Black lines indicate median. (T) CellChat analysis of TRM–neutrophil efferocytosis-associated signaling in young versus aged healthy donors. (Top left) Number of interactions. (Top right) Interaction (Int.) strength. (Bottom) Δ Strength (Young – Aged) for the top differentially altered signaling pathways; pathway abbreviations, key molecules, and efferocytosis roles are provided in table S1. (U) to (W) As in (R) to (T) but comparing healthy versus diseased donors. Neutrophil subclustering identifies five subclusters (Neu1 to 5) (U), with senescence module scores comparing Neu1+5 (healthy-enriched) versus Neu2+3+4 (diseased-enriched) (V), and CellChat analysis of healthy versus diseased TRM–neutrophil communication (W).

this functional reserve, consistent with CCA programs in a noncanonical context. We link these transcriptomic features to surface CXCR4 expression, enabling prospective identification by flow cytometry, and demonstrate a senescence-to-apoptosis transition (Neu2 to Neu3) that accounts for CXCR4 and annexin V copositivity in aging tissues. Together, these findings extend current models of senescence-related immune dysfunction and suggest that targeting these dysfunctional neutrophil states may have therapeutic relevance.

Aging broadly impaired TRM efferocytosis of multiple cellular substrates, including senescent neutrophils and apoptotic thymocytes, lymphocytes, and erythrocytes, indicating a generalized decline in TRM clearance machinery. However, impairment was disproportionately severe for senescent neutrophils, reflecting their additional requirement for integrin-mediated stabilization beyond shared recognition defects. This two-hit vulnerability positions senescent neutrophils as a particularly pathogenic substrate and suggests that targeting their clearance may yield outsized therapeutic benefit.

When clearance fails, senescent neutrophils exert paracrine stress on surrounding parenchymal cells. In the liver, hepatocytes up-regulated ICAM-1, sinusoidal endothelial cells increased VCAM-1, and stellate cells deposited fibronectin in close spatial proximity to CXCR4⁺ neutrophils. Thus, impaired efferocytosis generates a local inflammatory microenvironment that propagates dysfunction beyond the immune compartment, linking neutrophil accumulation to inflammatory, vascular, and fibrotic features of tissue aging in a feed-forward cycle.

Mechanistically, integrin-mediated stabilization emerged as the dominant disrupted step. Although LFA-1 was counteractively up-regulated across all three efferocytic organs, transcriptional suppression of key conformational activators (*Tln2*, *Fermt3*, *Skap1*, *Rac1*, *RhoA*) locked integrins in a low-affinity state. Downstream post-engulfment programs were similarly suppressed through EP2-driven TF dysregulation, with reduced pro-phagocytic JUNB, STAT3, and RELA and enhanced anti-phagocytic CREB1. EP2 signaling therefore acts as a central regulator of efferocytosis by impairing both integrin activation and downstream cargo processing.

Pharmacological EP2 inhibition demonstrated the reversibility of defective TRM efferocytosis. PF-04418948 treatment reduced senescent neutrophil burden across the liver, spleen, and bone marrow and restored KC efferocytosis ex vivo. Rescue was most complete in the liver, potentially due to greater drug accessibility through the portal circulation, whereas the spleen and bone marrow may require longer or earlier inhibition.

In the aged and diseased human liver, TRM proportions declined, *PTGER2* expression was selectively elevated in TRMs, neutrophils shifted toward senescent-like transcriptional states, and TRM–neutrophil communication was reduced, paralleling our findings in mice. The aged human heart similarly exhibited TRM loss and selective *PTGER2* elevation, pointing to a conserved change in TRM biology across organs in aging. Although *PTGER2* did not reach genome-wide significance in longevity GWAS, its association with age-related diseases, cancer, and immune dysfunction underscores its translational relevance (94) and highlights limitations of current GWAS in capturing cell-state-dependent mechanisms. These human findings, while correlative, position the TRM EP2–efferocytosis axis as a candidate mechanism in human aging that warrants further functional testing. Specifically, future studies should assess whether the impaired clearance of senescent neutrophils also occurs in human TRMs and whether pharmacological EP2 blockade can restore this defect.

Several limitations merit consideration. Our flow cytometric analyses sampled selected ligand-receptor pairs, and would not have captured all efferocytic axes, including MerTK–Gas6 signaling in liver, which can be functionally impaired by proteolytic cleavage without reduced surface abundance (63). Human scRNA-seq analyses were constrained by the limited availability of well-powered, age-stratified datasets.

Future studies should test whether disrupting neutrophil-driven paracrine stress prevents parenchymal injury across additional organs and whether EP2 antagonism might synergize with senolytic or senomorphic therapies. Given the outsized role of TRMs in driving organ-wide inflammation and functional decline, restoring TRM health, for example through pharmacologic EP2 inhibition, may limit the development of

age-associated disorders including frailty and sarcopenia, cardiovascular and liver disease, and neurodegeneration, with CXCR4⁺ neutrophil burden serving as a potential pharmacodynamic biomarker.

Methods summary

A full description of the methods can be found in the supplementary materials. The supplementary materials and methods section includes descriptions of mouse models and behavioral assessments (Barnes maze, novel object recognition); frailty and grip strength evaluation; cardiac imaging and histology (MRI, TEM, trichrome staining); Luminex cytokine profiling and TMT-based plasma proteomics; serum liver biomarker measurements; single-cell RNA-seq library preparation and bioinformatic analysis of mouse liver; Western blot; peritoneal macrophage isolation and mitochondrial function assessment by flow cytometry; multiorgan flow cytometry for macrophage and neutrophil characterization; ex vivo efferocytosis assays; immunohistochemistry of muscle, liver, and spleen tissues; MACSima multiplex immunofluorescence imaging and QuPath-based quantification; reanalysis of published human liver and heart scRNA-seq datasets; MAGMA-based human longevity GWAS analysis; and statistical methods.

REFERENCES AND NOTES

- C. López-Otin, M. A. Blasco, L. Partridge, M. Serrano, G. Kroemer, The hallmarks of aging. *Cell* **153**, 1194–1217 (2013). doi: [10.1016/j.cell.2013.05.039](https://doi.org/10.1016/j.cell.2013.05.039); pmid: [23746838](https://pubmed.ncbi.nlm.nih.gov/23746838/)
- B. K. Kennedy *et al.*, Geroscience: Linking aging to chronic disease. *Cell* **159**, 709–713 (2014). doi: [10.1016/j.cell.2014.10.039](https://doi.org/10.1016/j.cell.2014.10.039); pmid: [25417146](https://pubmed.ncbi.nlm.nih.gov/25417146/)
- N. Schaum *et al.*, Ageing hallmarks exhibit organ-specific temporal signatures. *Nature* **583**, 596–602 (2020). doi: [10.1038/s41586-020-2499-y](https://doi.org/10.1038/s41586-020-2499-y); pmid: [32669715](https://pubmed.ncbi.nlm.nih.gov/32669715/)
- T. A. Wynn, A. Chawla, J. W. Pollard, Macrophage biology in development, homeostasis and disease. *Nature* **496**, 445–455 (2013). doi: [10.1038/nature12034](https://doi.org/10.1038/nature12034); pmid: [23619691](https://pubmed.ncbi.nlm.nih.gov/23619691/)
- E. Linehan, D. C. Fitzgerald, Ageing and the immune system: Focus on macrophages. *Eur. J. Microbiol. Immunol.* **5**, 14–24 (2015). doi: [10.1556/EuJMI-D-14-00035](https://doi.org/10.1556/EuJMI-D-14-00035); pmid: [25883791](https://pubmed.ncbi.nlm.nih.gov/25883791/)
- F. Ginhoux, M. Williams, Tissue-Resident Macrophage Ontogeny and Homeostasis. *Immunity* **44**, 439–449 (2016). doi: [10.1016/j.immuni.2016.02.024](https://doi.org/10.1016/j.immuni.2016.02.024); pmid: [26982352](https://pubmed.ncbi.nlm.nih.gov/26982352/)
- D. Hashimoto *et al.*, Tissue-resident macrophages self-maintain locally throughout adult life with minimal contribution from circulating monocytes. *Immunity* **38**, 792–804 (2013). doi: [10.1016/j.immuni.2013.04.004](https://doi.org/10.1016/j.immuni.2013.04.004); pmid: [23601688](https://pubmed.ncbi.nlm.nih.gov/23601688/)
- S. A. Dick *et al.*, Three tissue resident macrophage subsets coexist across organs with conserved origins and life cycles. *Sci. Immunol.* **7**, eabf7777 (2022). doi: [10.1126/sciimmunol.abf7777](https://doi.org/10.1126/sciimmunol.abf7777); pmid: [34995099](https://pubmed.ncbi.nlm.nih.gov/34995099/)
- E. Mass *et al.*, Specification of tissue-resident macrophages during organogenesis. *Science* **353**, aaf4238 (2016). doi: [10.1126/science.aaf4238](https://doi.org/10.1126/science.aaf4238); pmid: [27492475](https://pubmed.ncbi.nlm.nih.gov/27492475/)
- G. Hoeffel, F. Ginhoux, Fetal monocytes and the origins of tissue-resident macrophages. *Cell. Immunol.* **330**, 5–15 (2018). doi: [10.1016/j.cellimm.2018.01.001](https://doi.org/10.1016/j.cellimm.2018.01.001); pmid: [29475558](https://pubmed.ncbi.nlm.nih.gov/29475558/)
- C. C. Bain *et al.*, Constant replenishment from circulating monocytes maintains the macrophage pool in the intestine of adult mice. *Nat. Immunol.* **15**, 929–937 (2014). doi: [10.1038/ni.2967](https://doi.org/10.1038/ni.2967); pmid: [25151491](https://pubmed.ncbi.nlm.nih.gov/25151491/)
- S. De Schepper *et al.*, Self-Maintaining Gut Macrophages Are Essential for Intestinal Homeostasis. *Cell* **175**, 400–415.e13 (2018). doi: [10.1016/j.cell.2018.07.048](https://doi.org/10.1016/j.cell.2018.07.048); pmid: [30173915](https://pubmed.ncbi.nlm.nih.gov/30173915/)
- E. L. Gautier *et al.*, Gene-expression profiles and transcriptional regulatory pathways that underlie the identity and diversity of mouse tissue macrophages. *Nat. Immunol.* **13**, 1118–1128 (2012). doi: [10.1038/ni.2419](https://doi.org/10.1038/ni.2419); pmid: [23023392](https://pubmed.ncbi.nlm.nih.gov/23023392/)
- E. G. Perdiguero, F. Geissmann, The development and maintenance of resident macrophages. *Nat. Immunol.* **17**, 2–8 (2016). doi: [10.1038/ni.3341](https://doi.org/10.1038/ni.3341); pmid: [26681456](https://pubmed.ncbi.nlm.nih.gov/26681456/)
- T. L. Tay, J. C. Savage, C. W. Hui, K. Bisht, M. E. Tremblay, Microglia across the lifespan: From origin to function in brain development, plasticity and cognition. *J. Physiol.* **595**, 1929–1945 (2017). doi: [10.1113/JP272134](https://doi.org/10.1113/JP272134); pmid: [27104646](https://pubmed.ncbi.nlm.nih.gov/27104646/)
- K. Molawi *et al.*, Progressive replacement of embryo-derived cardiac macrophages with age. *J. Exp. Med.* **211**, 2151–2158 (2014). doi: [10.1084/jem.20140639](https://doi.org/10.1084/jem.20140639); pmid: [25245760](https://pubmed.ncbi.nlm.nih.gov/25245760/)
- S. H. Lee, P. M. Starkey, S. Gordon, Quantitative analysis of total macrophage content in adult mouse tissues. Immunochemical studies with monoclonal antibody F4/80. *J. Exp. Med.* **161**, 475–489 (1985). doi: [10.1084/jem.161.3.475](https://doi.org/10.1084/jem.161.3.475); pmid: [3973536](https://pubmed.ncbi.nlm.nih.gov/3973536/)
- G. R. Lee, J. Foerster, J. W. Athens, J. L. Lukens, *Wintrobe's Clinical Hematology* (Lea & Febiger, ed. 9, 1993).
- E. Boada-Romero, J. Martinez, B. L. Heckmann, D. R. Green, The clearance of dead cells by efferocytosis. *Nat. Rev. Mol. Cell Biol.* **21**, 398–414 (2020). doi: [10.1038/s41580-020-0232-1](https://doi.org/10.1038/s41580-020-0232-1); pmid: [32251387](https://pubmed.ncbi.nlm.nih.gov/32251387/)
- J. T. Dancy, K. A. Deubelbeiss, L. A. Harker, C. A. Finch, Neutrophil kinetics in man. *J. Clin. Invest.* **58**, 705–715 (1976). doi: [10.1172/JCI108517](https://doi.org/10.1172/JCI108517); pmid: [956397](https://pubmed.ncbi.nlm.nih.gov/956397/)
- C. Summers *et al.*, Neutrophil kinetics in health and disease. *Trends Immunol.* **31**, 318–324 (2010). doi: [10.1016/j.it.2010.05.006](https://doi.org/10.1016/j.it.2010.05.006); pmid: [20620114](https://pubmed.ncbi.nlm.nih.gov/20620114/)
- M. Casanova-Acebes *et al.*, Rhythmic modulation of the hematopoietic niche through neutrophil clearance. *Cell* **153**, 1025–1035 (2013). doi: [10.1016/j.cell.2013.04.040](https://doi.org/10.1016/j.cell.2013.04.040); pmid: [23706740](https://pubmed.ncbi.nlm.nih.gov/23706740/)
- A. Hidalgo, E. R. Chilvers, C. Summers, L. Koenderman, The Neutrophil Life Cycle. *Trends Immunol.* **40**, 584–597 (2019). doi: [10.1016/j.it.2019.04.013](https://doi.org/10.1016/j.it.2019.04.013); pmid: [31153737](https://pubmed.ncbi.nlm.nih.gov/31153737/)
- E. Kolaczowska, P. Kubes, Neutrophil recruitment and function in health and inflammation. *Nat. Rev. Immunol.* **13**, 159–175 (2013). doi: [10.1038/nri3399](https://doi.org/10.1038/nri3399); pmid: [23435331](https://pubmed.ncbi.nlm.nih.gov/23435331/)
- N. Jarzewska *et al.*, Neutrophil Extracellular Traps (NETs) as a Potential Target for Anti-Aging: Role of Therapeutic Apheresis. *Horm. Metab. Res.* **57**, 632–638 (2025). doi: [10.1055/a-2444-3422](https://doi.org/10.1055/a-2444-3422); pmid: [39788160](https://pubmed.ncbi.nlm.nih.gov/39788160/)
- A. Lagnado *et al.*, Neutrophils induce paracrine telomere dysfunction and senescence in ROS-dependent manner. *EMBO J.* **40**, e106048 (2021). doi: [10.15252/emboj.2020106048](https://doi.org/10.15252/emboj.2020106048); pmid: [33764576](https://pubmed.ncbi.nlm.nih.gov/33764576/)
- R. C. Furze, S. M. Rankin, The role of the bone marrow in neutrophil clearance under homeostatic conditions in the mouse. *FASEB J.* **22**, 3111–3119 (2008). doi: [10.1096/fj.08-109876](https://doi.org/10.1096/fj.08-109876); pmid: [18509199](https://pubmed.ncbi.nlm.nih.gov/18509199/)
- C. Hong *et al.*, Coordinate regulation of neutrophil homeostasis by liver X receptors in mice. *J. Clin. Invest.* **122**, 337–347 (2012). doi: [10.1172/JCI58393](https://doi.org/10.1172/JCI58393); pmid: [22156197](https://pubmed.ncbi.nlm.nih.gov/22156197/)
- E. Knudsen, H. B. Benestad, T. Seierstad, P. O. Iversen, Macrophages in spleen and liver direct the migration pattern of rat neutrophils during inflammation. *Eur. J. Haematol.* **73**, 109–122 (2004). doi: [10.1111/j.1600-0609.2004.00263.x](https://doi.org/10.1111/j.1600-0609.2004.00263.x); pmid: [15245510](https://pubmed.ncbi.nlm.nih.gov/15245510/)
- A. Singhal, S. Kumar, Neutrophil and remnant clearance in immunity and inflammation. *Immunology* **165**, 22–43 (2022). doi: [10.1111/imm.13423](https://doi.org/10.1111/imm.13423); pmid: [34704249](https://pubmed.ncbi.nlm.nih.gov/34704249/)
- P. S. Minhas *et al.*, Restoring metabolism of myeloid cells reverses cognitive decline in ageing. *Nature* **590**, 122–128 (2021). doi: [10.1038/s41586-020-03160-0](https://doi.org/10.1038/s41586-020-03160-0); pmid: [33473210](https://pubmed.ncbi.nlm.nih.gov/33473210/)
- S. H. Cheshier, S. J. Morrison, X. Liao, I. L. Weissman, In vivo proliferation and cell cycle kinetics of long-term self-renewing hematopoietic stem cells. *Proc. Natl. Acad. Sci. U.S.A.* **96**, 3120–3125 (1999). doi: [10.1073/pnas.96.6.3120](https://doi.org/10.1073/pnas.96.6.3120); pmid: [10077647](https://pubmed.ncbi.nlm.nih.gov/10077647/)
- A. McCabe, K. C. MacNamara, Macrophages: Key regulators of steady-state and demand-adapted hematopoiesis. *Exp. Hematol.* **44**, 213–222 (2016). doi: [10.1016/j.exphem.2016.01.003](https://doi.org/10.1016/j.exphem.2016.01.003); pmid: [26806720](https://pubmed.ncbi.nlm.nih.gov/26806720/)
- C. L. Scott *et al.*, Bone marrow-derived monocytes give rise to self-renewing and fully differentiated KCs. *Nat. Commun.* **7**, 10321 (2016). doi: [10.1038/ncomms10321](https://doi.org/10.1038/ncomms10321); pmid: [26813785](https://pubmed.ncbi.nlm.nih.gov/26813785/)
- T. N. Shaw *et al.*, Tissue-resident macrophages in the intestine are long lived and defined by Tim-4 and CD4 expression. *J. Exp. Med.* **215**, 1507–1518 (2018). doi: [10.1084/jem.20180019](https://doi.org/10.1084/jem.20180019); pmid: [29789388](https://pubmed.ncbi.nlm.nih.gov/29789388/)
- H. Keren-Shaul *et al.*, A Unique Microglia Type Associated with Restricting Development of Alzheimer's Disease. *Cell* **169**, 1276–1290.e17 (2017). doi: [10.1016/j.cell.2017.05.018](https://doi.org/10.1016/j.cell.2017.05.018); pmid: [28602351](https://pubmed.ncbi.nlm.nih.gov/28602351/)
- H. Van Hove *et al.*, A single-cell atlas of mouse brain macrophages reveals unique transcriptional identities shaped by ontogeny and tissue environment. *Nat. Neurosci.* **22**, 1021–1035 (2019). doi: [10.1038/s41593-019-0393-4](https://doi.org/10.1038/s41593-019-0393-4); pmid: [31061494](https://pubmed.ncbi.nlm.nih.gov/31061494/)
- K. Alves de Lima, J. Rustenhoven, J. Kipnis, Meningeal Immunity and Its Function in Maintenance of the Central Nervous System in Health and Disease. *Annu. Rev. Immunol.* **38**, 597–620 (2020). doi: [10.1146/annurev-immunol-102319-103410](https://doi.org/10.1146/annurev-immunol-102319-103410); pmid: [32340575](https://pubmed.ncbi.nlm.nih.gov/32340575/)
- Ados. A. Cassado, M. R. D'Império Lima, K. R. Bortoluci, Revisiting mouse peritoneal macrophages: Heterogeneity, development, and function. *Front. Immunol.* **6**, 225 (2015). doi: [10.3389/fimmu.2015.00225](https://doi.org/10.3389/fimmu.2015.00225); pmid: [26042120](https://pubmed.ncbi.nlm.nih.gov/26042120/)
- P. Li, Z. Fan, Y. Huang, L. Luo, X. Wu, Mitochondrial dynamics at the intersection of macrophage polarization and metabolism. *Front. Immunol.* **16**, 1520814 (2025). doi: [10.3389/fimmu.2025.1520814](https://doi.org/10.3389/fimmu.2025.1520814); pmid: [40196123](https://pubmed.ncbi.nlm.nih.gov/40196123/)
- F. Ginhoux *et al.*, Fate mapping analysis reveals that adult microglia derive from primitive macrophages. *Science* **330**, 841–845 (2010). doi: [10.1126/science.1194637](https://doi.org/10.1126/science.1194637); pmid: [20966214](https://pubmed.ncbi.nlm.nih.gov/20966214/)
- S. Schiaffino, C. Reggiani, Fiber types in mammalian skeletal muscles. *Physiol. Rev.* **91**, 1447–1531 (2011). doi: [10.1152/physrev.00031.2010](https://doi.org/10.1152/physrev.00031.2010); pmid: [22013216](https://pubmed.ncbi.nlm.nih.gov/22013216/)
- M. Azevedo, M. K. Baylies, Getting into Position: Nuclear Movement in Muscle Cells. *Trends Cell Biol.* **30**, 303–316 (2020). doi: [10.1016/j.tcb.2020.01.002](https://doi.org/10.1016/j.tcb.2020.01.002); pmid: [32008895](https://pubmed.ncbi.nlm.nih.gov/32008895/)
- D. F. Dai, P. S. Rabinovitch, Z. Ungvari, Mitochondria and cardiovascular aging. *Circ. Res.* **110**, 1109–1124 (2012). doi: [10.1161/CIRCRESAHA.111.246140](https://doi.org/10.1161/CIRCRESAHA.111.246140); pmid: [22499901](https://pubmed.ncbi.nlm.nih.gov/22499901/)
- V. Ronca *et al.*, The liver as a central “hub” of the immune system: Pathophysiological implications. *Physiol. Rev.* **105**, 493–539 (2025). doi: [10.1152/physrev.00004.2023](https://doi.org/10.1152/physrev.00004.2023); pmid: [39297676](https://pubmed.ncbi.nlm.nih.gov/39297676/)
- M. Ogródnik *et al.*, Guidelines for minimal information on cellular senescence experimentation in vivo. *Cell* **187**, 4150–4175 (2024). doi: [10.1016/j.cell.2024.05.059](https://doi.org/10.1016/j.cell.2024.05.059); pmid: [39121846](https://pubmed.ncbi.nlm.nih.gov/39121846/)
- D. Saul *et al.*, A new gene set identifies senescent cells and predicts senescence-associated pathways across tissues. *Nat. Commun.* **13**, 4827 (2022). doi: [10.1038/s41467-022-32552-1](https://doi.org/10.1038/s41467-022-32552-1); pmid: [35974106](https://pubmed.ncbi.nlm.nih.gov/35974106/)

48. V. Suryadevara *et al.*, SenNet recommendations for detecting senescent cells in different tissues. *Nat. Rev. Mol. Cell Biol.* **25**, 1001–1023 (2024). doi: [10.1038/s41580-024-00738-8](https://doi.org/10.1038/s41580-024-00738-8); pmid: [38831121](https://pubmed.ncbi.nlm.nih.gov/38831121/)
49. B. Amulic *et al.*, Cell-Cycle Proteins Control Production of Neutrophil Extracellular Traps. *Dev. Cell* **43**, 449–462.e5 (2017). doi: [10.1016/j.devcel.2017.10.013](https://doi.org/10.1016/j.devcel.2017.10.013); pmid: [29103955](https://pubmed.ncbi.nlm.nih.gov/29103955/)
50. P. Klausen, M. D. Bjerregaard, N. Borregaard, J. B. Cowland, End-stage differentiation of neutrophil granulocytes in vivo is accompanied by up-regulation of p27kip1 and down-regulation of CDK2, CDK4, and CDK6. *J. Leukoc. Biol.* **75**, 569–578 (2004). doi: [10.1189/jlb.1003474](https://doi.org/10.1189/jlb.1003474); pmid: [14694185](https://pubmed.ncbi.nlm.nih.gov/14694185/)
51. R. Syahirah, A. Y. Hsu, Q. Deng, A curious case of cyclin-dependent kinases in neutrophils. *J. Leukoc. Biol.* **111**, 1057–1068 (2022). doi: [10.1002/JLB.2RU1021-573R](https://doi.org/10.1002/JLB.2RU1021-573R); pmid: [35188696](https://pubmed.ncbi.nlm.nih.gov/35188696/)
52. C. Rosales, Neutrophil: A Cell with Many Roles in Inflammation or Several Cell Types? *Front. Physiol.* **9**, 113 (2018). doi: [10.3389/fphys.2018.00113](https://doi.org/10.3389/fphys.2018.00113); pmid: [29515456](https://pubmed.ncbi.nlm.nih.gov/29515456/)
53. F. Zhang *et al.*, Neutrophil diversity and function in health and disease. *Signal Transduct. Target. Ther.* **9**, 343 (2024). doi: [10.1038/s41392-024-02049-y](https://doi.org/10.1038/s41392-024-02049-y); pmid: [39638788](https://pubmed.ncbi.nlm.nih.gov/39638788/)
54. K. De Filippo, S. M. Rankin, CXCR4, the master regulator of neutrophil trafficking in homeostasis and disease. *Eur. J. Clin. Invest.* **48**, e12949 (2018). doi: [10.1111/eci.12949](https://doi.org/10.1111/eci.12949); pmid: [29734477](https://pubmed.ncbi.nlm.nih.gov/29734477/)
55. A. C. Doran, A. Yurdagül Jr., I. Tabas, Efferocytosis in health and disease. *Nat. Rev. Immunol.* **20**, 254–267 (2020). doi: [10.1038/s41577-019-0240-6](https://doi.org/10.1038/s41577-019-0240-6); pmid: [31822793](https://pubmed.ncbi.nlm.nih.gov/31822793/)
56. S. Mylvaganam, S. A. Freeman, The resolution of phagosomes. *Immunol. Rev.* **319**, 45–64 (2023). doi: [10.1111/imr.13260](https://doi.org/10.1111/imr.13260); pmid: [37551912](https://pubmed.ncbi.nlm.nih.gov/37551912/)
57. B. Moon, S. Yang, H. Moon, J. Lee, D. Park, After cell death: The molecular machinery of efferocytosis. *Exp. Mol. Med.* **55**, 1644–1651 (2023). doi: [10.1038/s12276-023-01070-5](https://doi.org/10.1038/s12276-023-01070-5); pmid: [37612408](https://pubmed.ncbi.nlm.nih.gov/37612408/)
58. S. Toda, K. Segawa, S. Nagata, MerTK-mediated engulfment of pyrenocytes by central macrophages in erythroblastic islands. *Blood* **123**, 3963–3971 (2014). doi: [10.1182/blood-2014-01-547976](https://doi.org/10.1182/blood-2014-01-547976); pmid: [24659633](https://pubmed.ncbi.nlm.nih.gov/24659633/)
59. V. A. Fadok *et al.*, Exposure of phosphatidylserine on the surface of apoptotic lymphocytes triggers specific recognition and removal by macrophages. *J. Immunol.* **148**, 2207–2216 (1992). doi: [10.4049/jimmunol.148.7.2207](https://doi.org/10.4049/jimmunol.148.7.2207); pmid: [1545126](https://pubmed.ncbi.nlm.nih.gov/1545126/)
60. R. E. Mebius, G. Kraal, Structure and function of the spleen. *Nat. Rev. Immunol.* **5**, 606–616 (2005). doi: [10.1038/nri1669](https://doi.org/10.1038/nri1669); pmid: [16056254](https://pubmed.ncbi.nlm.nih.gov/16056254/)
61. R. Swan *et al.*, Polymicrobial sepsis enhances clearance of apoptotic immune cells by splenic macrophages. *Surgery* **142**, 253–261 (2007). doi: [10.1016/j.surg.2007.04.005](https://doi.org/10.1016/j.surg.2007.04.005); pmid: [17689693](https://pubmed.ncbi.nlm.nih.gov/17689693/)
62. M. Kawano, S. Nagata, Lupus-like autoimmune disease caused by a lack of Xkr8, a caspase-dependent phospholipid scramblase. *Proc. Natl. Acad. Sci. U.S.A.* **115**, 2132–2137 (2018). doi: [10.1073/pnas.1720732115](https://doi.org/10.1073/pnas.1720732115); pmid: [29440417](https://pubmed.ncbi.nlm.nih.gov/29440417/)
63. H. Hu *et al.*, Defective efferocytosis by aged macrophages promotes STING signaling mediated inflammatory liver injury. *Cell Death Discov.* **9**, 236 (2023). doi: [10.1038/s41420-023-01497-9](https://doi.org/10.1038/s41420-023-01497-9); pmid: [37422464](https://pubmed.ncbi.nlm.nih.gov/37422464/)
64. A. Kinkhabwala *et al.*, MACSima imaging cyclic staining (MICS) technology reveals combinatorial target pairs for CAR T cell treatment of solid tumors. *Sci. Rep.* **12**, 1911 (2022). doi: [10.1038/s41598-022-05841-4](https://doi.org/10.1038/s41598-022-05841-4); pmid: [35115587](https://pubmed.ncbi.nlm.nih.gov/35115587/)
65. A. S. O'Neill, T. K. van den Berg, G. E. Mullen, Sialoadhesin - a macrophage-restricted marker of immunoregulation and inflammation. *Immunology* **138**, 198–207 (2013). doi: [10.1111/imm.12042](https://doi.org/10.1111/imm.12042); pmid: [23181380](https://pubmed.ncbi.nlm.nih.gov/23181380/)
66. V. Papayannopoulos, Neutrophil extracellular traps in immunity and disease. *Nat. Rev. Immunol.* **18**, 134–147 (2018). doi: [10.1038/nri.2017.105](https://doi.org/10.1038/nri.2017.105); pmid: [28990587](https://pubmed.ncbi.nlm.nih.gov/28990587/)
67. M. Williams *et al.*, Spatial proteogenomics reveals distinct and evolutionarily conserved hepatic macrophage niches. *Cell* **185**, 379–396.e38 (2022). doi: [10.1016/j.cell.2021.12.018](https://doi.org/10.1016/j.cell.2021.12.018); pmid: [35021063](https://pubmed.ncbi.nlm.nih.gov/35021063/)
68. J. C. Acosta *et al.*, A complex secretory program orchestrated by the inflammasome controls paracrine senescence. *Nat. Cell Biol.* **15**, 978–990 (2013). doi: [10.1038/ncb2784](https://doi.org/10.1038/ncb2784); pmid: [23770676](https://pubmed.ncbi.nlm.nih.gov/23770676/)
69. M. V. Gulubova, Intercellular adhesion molecule-1 (ICAM-1) expression in the liver of patients with extrahepatic cholestasis. *Acta Histochem.* **100**, 59–74 (1998). doi: [10.1016/S0065-1281\(98\)80006-8](https://doi.org/10.1016/S0065-1281(98)80006-8); pmid: [9542581](https://pubmed.ncbi.nlm.nih.gov/9542581/)
70. J. M. Cook-Mills, M. E. Marchese, H. Abdala-Valencia, Vascular cell adhesion molecule-1 expression and signaling during disease: Regulation by reactive oxygen species and antioxidants. *Antioxid. Redox Signal.* **15**, 1607–1638 (2011). doi: [10.1089/ars.2010.3522](https://doi.org/10.1089/ars.2010.3522); pmid: [21050132](https://pubmed.ncbi.nlm.nih.gov/21050132/)
71. A. Puente *et al.*, LOXL2-A New Target in Antifibrogenic Therapy? *Int. J. Mol. Sci.* **20**, 1634 (2019). doi: [10.3390/ijms20071634](https://doi.org/10.3390/ijms20071634); pmid: [30986934](https://pubmed.ncbi.nlm.nih.gov/30986934/)
72. S. Jin *et al.*, Inference and analysis of cell-cell communication using CellChat. *Nat. Commun.* **12**, 1088 (2021). doi: [10.1038/s41467-021-21246-9](https://doi.org/10.1038/s41467-021-21246-9); pmid: [33597522](https://pubmed.ncbi.nlm.nih.gov/33597522/)
73. M. Bednarczyk, H. Stege, S. Grabbe, M. Bros, β 2 Integrins-Multi-Functional Leukocyte Receptors in Health and Disease. *Int. J. Mol. Sci.* **21**, 1402 (2020). doi: [10.3390/ijms21041402](https://doi.org/10.3390/ijms21041402); pmid: [32092981](https://pubmed.ncbi.nlm.nih.gov/32092981/)
74. H. Sun, K. Zhi, L. Hu, Z. Fan, The Activation and Regulation of β 2 Integrins in Phagocytes and Phagocytosis. *Front. Immunol.* **12**, 633639 (2021). doi: [10.3389/fimmu.2021.633639](https://doi.org/10.3389/fimmu.2021.633639); pmid: [33868253](https://pubmed.ncbi.nlm.nih.gov/33868253/)
75. A. Torres-Gomez, C. Cabañas, E. M. Lafuente, Phagocytic Integrins: Activation and Signaling. *Front. Immunol.* **11**, 738 (2020). doi: [10.3389/fimmu.2020.00738](https://doi.org/10.3389/fimmu.2020.00738); pmid: [32425937](https://pubmed.ncbi.nlm.nih.gov/32425937/)
76. P. Angel, M. Karin, The role of Jun, Fos and the AP-1 complex in cell-proliferation and transformation. *Biochim. Biophys. Acta* **1072**, 129–157 (1991). pmid: [1751545](https://pubmed.ncbi.nlm.nih.gov/1751545/)
77. C. Hong, P. Tontonoz, Coordination of inflammation and metabolism by PPAR and LXR nuclear receptors. *Curr. Opin. Genet. Dev.* **18**, 461–467 (2008). doi: [10.1016/j.gde.2008.07.016](https://doi.org/10.1016/j.gde.2008.07.016); pmid: [18782619](https://pubmed.ncbi.nlm.nih.gov/18782619/)
78. C. Cavelier, I. Lorenzi, L. Rohrer, A. von Eckardstein, Lipid efflux by the ATP-binding cassette transporters ABCA1 and ABCG1. *Biochim. Biophys. Acta* **1761**, 655–666 (2006). doi: [10.1016/j.bbalip.2006.04.012](https://doi.org/10.1016/j.bbalip.2006.04.012); pmid: [16798073](https://pubmed.ncbi.nlm.nih.gov/16798073/)
79. L. Campana *et al.*, The STAT3-IL-10-IL-6 Pathway Is a Novel Regulator of Macrophage Efferocytosis and Phenotypic Conversion in Sterile Liver Injury. *J. Immunol.* **200**, 1169–1187 (2018). doi: [10.4049/jimmunol.1701247](https://doi.org/10.4049/jimmunol.1701247); pmid: [29263216](https://pubmed.ncbi.nlm.nih.gov/29263216/)
80. T. Lawrence, The nuclear factor NF- κ B pathway in inflammation. *Cold Spring Harb. Perspect. Biol.* **1**, a001651 (2009). doi: [10.1101/cshperspect.a001651](https://doi.org/10.1101/cshperspect.a001651); pmid: [20457564](https://pubmed.ncbi.nlm.nih.gov/20457564/)
81. H. Cheon *et al.*, Prostaglandin E2 augments IL-10 signaling and function. *J. Immunol.* **177**, 1092–1100 (2006). doi: [10.4049/jimmunol.177.2.1092](https://doi.org/10.4049/jimmunol.177.2.1092); pmid: [16818766](https://pubmed.ncbi.nlm.nih.gov/16818766/)
82. W. H. Faour *et al.*, Early growth response factor-1 mediates prostaglandin E2-dependent transcriptional suppression of cytokine-induced tumor necrosis factor- α gene expression in human macrophages and rheumatoid arthritis-affected synovial fibroblasts. *J. Biol. Chem.* **280**, 9536–9546 (2005). doi: [10.1074/jbc.M414067200](https://doi.org/10.1074/jbc.M414067200); pmid: [15640148](https://pubmed.ncbi.nlm.nih.gov/15640148/)
83. E. J. Lee *et al.*, 15-Keto prostaglandin E₂ suppresses STAT3 signaling and inhibits breast cancer cell growth and progression. *Redox Biol.* **23**, 101175 (2019). doi: [10.1016/j.redox.2019.101175](https://doi.org/10.1016/j.redox.2019.101175); pmid: [31129031](https://pubmed.ncbi.nlm.nih.gov/31129031/)
84. F. J. Ren, X. Y. Cai, Y. Yao, G. Y. Fang, JunB: A paradigm for Jun family in immune response and cancer. *Front. Cell. Infect. Microbiol.* **13**, 1222265 (2023). doi: [10.3389/fcimb.2023.1222265](https://doi.org/10.3389/fcimb.2023.1222265); pmid: [37731821](https://pubmed.ncbi.nlm.nih.gov/37731821/)
85. Y. P. Zhu, J. R. Brown, D. Sag, L. Zhang, J. Suttles, Adenosine 5'-monophosphate-activated protein kinase regulates IL-10-mediated anti-inflammatory signaling pathways in macrophages. *J. Immunol.* **194**, 584–594 (2015). doi: [10.4049/jimmunol.1401024](https://doi.org/10.4049/jimmunol.1401024); pmid: [25512602](https://pubmed.ncbi.nlm.nih.gov/25512602/)
86. K. J. af Forselles *et al.*, In vitro and in vivo characterization of PF-04418948, a novel, potent and selective prostaglandin EP₂ receptor antagonist. *Br. J. Pharmacol.* **164**, 1847–1856 (2011). doi: [10.1111/j.1476-5381.2011.01495.x](https://doi.org/10.1111/j.1476-5381.2011.01495.x); pmid: [21595651](https://pubmed.ncbi.nlm.nih.gov/21595651/)
87. X. Wu *et al.*, Macrophage WDFY3 mitigates autoimmunity by enhancing efferocytosis and suppressing T cell activation in mice. *Nat. Commun.* **16**, 8694 (2025). doi: [10.1038/s41467-025-63741-3](https://doi.org/10.1038/s41467-025-63741-3); pmid: [41027899](https://pubmed.ncbi.nlm.nih.gov/41027899/)
88. R. C. Jones *et al.*, The Tabula Sapiens: A multiple-organ, single-cell transcriptomic atlas of humans. *Science* **376**, eabl4896 (2022). doi: [10.1126/science.abl4896](https://doi.org/10.1126/science.abl4896); pmid: [35549404](https://pubmed.ncbi.nlm.nih.gov/35549404/)
89. P. R. H. J. Timmers, J. F. Wilson, P. K. Joshi, J. Deelen, Multivariate genomic scan implicates novel loci and haem metabolism in human ageing. *Nat. Commun.* **11**, 3570 (2020). doi: [10.1038/s41467-020-17312-3](https://doi.org/10.1038/s41467-020-17312-3); pmid: [32678081](https://pubmed.ncbi.nlm.nih.gov/32678081/)
90. A. Zenin *et al.*, Identification of 12 genetic loci associated with human healthspan. *Commun. Biol.* **2**, 41 (2019). doi: [10.1038/s42003-019-0290-0](https://doi.org/10.1038/s42003-019-0290-0); pmid: [30729179](https://pubmed.ncbi.nlm.nih.gov/30729179/)
91. L. C. Pilling *et al.*, Human longevity: 25 genetic loci associated in 389,166 UK biobank participants. *Aging* **9**, 2504–2520 (2017). doi: [10.18632/aging.101334](https://doi.org/10.18632/aging.101334); pmid: [29227965](https://pubmed.ncbi.nlm.nih.gov/29227965/)
92. P. R. Timmers *et al.*, Genomics of 1 million parent lifespans implicates novel pathways and common diseases and distinguishes survival chances. *eLife* **8**, e39856 (2019). doi: [10.7554/eLife.39856](https://doi.org/10.7554/eLife.39856); pmid: [30642433](https://pubmed.ncbi.nlm.nih.gov/30642433/)
93. J. Deelen *et al.*, A meta-analysis of genome-wide association studies identifies multiple longevity genes. *Nat. Commun.* **10**, 3669 (2019). doi: [10.1038/s41467-019-11558-2](https://doi.org/10.1038/s41467-019-11558-2); pmid: [31413261](https://pubmed.ncbi.nlm.nih.gov/31413261/)
94. J. Jiang, R. Dingleline, Prostaglandin receptor EP₂ in the crosshairs of anti-inflammation, anti-cancer, and neuroprotection. *Trends Pharmacol Sci* **34**, 413–423 (2013). doi: [10.1016/j.tips.2013.05.003](https://doi.org/10.1016/j.tips.2013.05.003); pmid: [23796953](https://pubmed.ncbi.nlm.nih.gov/23796953/)
95. Y. J. Tan, K. I. Andreasson, Restoring tissue-resident macrophage clearance of senescent neutrophils limits organ-wide aging. *Dryad* (2026); <https://doi.org/10.5061/dryad.0vt4b8hck>.

ACKNOWLEDGMENTS

We thank the Stanford Shared FACS Facility for instrument support and troubleshooting during flow cytometry acquisition; J. Perrino for TEM training and assistance; the Diagnostic Laboratory for experimental consultation; C. Wang for guidance on scRNA-seq library preparation and instrumentation; A. Seng at the Chan Zuckerberg Biohub Genomics Platform, UCSF, for advice on Illumina sequencing; J. Brook for suggesting the inclusion of lung tissue in immune cell analyses; and P. Minhas, H. Ennerfelt, and M. Xie for critical reading of the manuscript. **Funding:** This work was supported by NIH grants 1RF1AG080742, 1RF1AG070839, and P30 AG066515; the American Heart Association 19PABH1345800, the Phil and Penny Knight Initiative for Brain Resilience at the Wu Tsai Neurosciences Institute, Stanford University, and the Arc Institute Ignite Award (to K.I.A.); the Glenn Foundation for Medical Research Postdoctoral Fellowship in Aging Research, the 2025 Glenn Foundation for Medical Research Postdoctoral Fellowship Continuation Award, and the Wu Tsai Neurosciences Institute NPIL Pilot Award (to Y.J.T.). K.I.A. is a Chan Zuckerberg–

San Francisco Biohub Investigator. O.S. receives funding from the Deutsche Forschungsgemeinschaft (TRR332 projects A2 & Z1, S0876/16-1, CFU342 P1) and the Else Kröner Fresenius Stiftung (2023_EKSE.212). J.A.B. is supported by an HHMI Hanna Gray Fellowship. The Stanford Shared FACS Facility was supported by NIH 1S100D026831-01, S10RR025518-01, and the Parker Institute for Cancer Immunotherapy. The Cell Sciences Imaging Facility was supported by NIH 1S100D028536. The Wu Tsai Neurosciences Preclinical Imaging Laboratory was supported by NIH S100D025176. The MACSima platform was supported by the core facility Biology in Context (BIC) at the IZKF in Münster and the DFG (project 502158695).

Author contributions: Y.J.T., O.S., S.J.P., and K.I.A. designed the experiments. Y.J.T. coordinated all collaborative workflows, performed all experiments, data analysis, data interpretation, and figure generation unless otherwise noted. T.E.C. performed the Barnes maze, novel object recognition, grip strength, and muscle fiber staining experiments and analyzed the data; processed spleen and bone marrow samples for flow cytometry for PF-treated mice; processed spleen samples for efferocytosis assays in *Cx3cr1^{CreER}*-EP2 mice; and assisted with frailty scoring and PF-04418948 dosing. F.Y. processed mouse liver scRNA-seq data, processed human liver and heart scRNA-seq data, generated Seurat objects, and generated Fig. 4, A to D, and fig. S5, B and C. F.J.G.-M. processed TMT mass spectrometry raw data, performed quality control, and generated analyzed matrices under the supervision of S.J.P. D.E.A. and V.V.D. performed MACSima multiplex imaging data acquisition under the supervision of O.S. Q.W. maintained and bred *Cx3cr1^{CreER}*-EP2 and *Rosa26^{CreER}*-EP2 mouse colonies. A.B. assisted with TMT mass spectrometry sample preparation and loaded samples onto the mass spectrometer under the supervision of S.J.P. J.K. performed mouse MRI scanning and data acquisition. J.A.B. assisted with mouse liver scRNA-seq data quality control and performed human GWAS MAGMA analysis across six longevity datasets.

Y.J.T. and K.I.A. conceived and supervised the project, designed the experiments, interpreted the data, and wrote the manuscript. **Competing interests:** K.I.A. is a cofounder of Willow Neuroscience Inc. All other authors declare no competing interests. **Data, code, and materials availability:** The data that support the findings of this study are available in the main figures, supplementary figures and tables; additional data files are available at Dryad (95); The mouse liver scRNA-seq data have been deposited in the Gene Expression Omnibus (GEO) under accession number GSE319489. Published human liver scRNA-seq data were obtained from Williams *et al.* [(67); GEO: GSE192742]. Published human heart scRNA-seq data were obtained from the Tabula Sapiens Consortium (88). Human longevity GWAS summary statistics were obtained from the respective published studies 89–93). All custom R scripts for human data reanalysis and GWAS analysis are available on Dryad (67).

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SUPPLEMENTARY MATERIALS

science.org/doi/10.1126/science.aea3075

Materials and Methods; Figs. S1 to S12; Table S1; References (96–102); MDAR Reproducibility Checklist

Submitted 3 July 2025; accepted 11 May 2026

10.1126/science.aea3075



Restored clearance of senescent neutrophils by tissue-resident macrophages limits organ aging

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Science **393** (6808), eaea3075. DOI: 10.1126/science.aea3075

Editor's summary

Increased inflammation is associated with aging and is implicated in decreased function in aging tissues. Tan *et al.* explored changes in older tissue-resident macrophages and investigated how these cells may contribute to aging phenotypes in mice (see the Perspective by FitzGerald). They found that signaling by the lipid messenger prostaglandin E2 was increased in these cells. In various mouse tissues, preventing an increase in tissue-resident macrophages helped to maintain mitochondrial function and limited other deleterious effects of aging. Tissue-resident macrophages help to reduce inflammation by removing apoptotic and damaged cells. One consequence of inhibiting prostaglandin E2 signaling was restored clearance of senescent neutrophils. These results emphasize the possible roles of macrophages and neutrophil clearance in tissue dysfunction during aging. —L. Bryan Ray

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